Requirement Specification & Design Document

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I. Functional Requirements (FR) – Quiz Examination System

FR-01: User Authentication – Register / Login

-Description: The system must allow users to register an account and log in to access the system.

-Acceptance Criteria:

Users can enter Username, Email, and Password.

Registration is successful when data is valid.

Login is successful with an existing account.

Display error messages if information is incorrect.

-Business Rules:

BR-01: Username must be unique.

BR-02: Email must be in the correct format and unique.

BR-03: Password must be at least 8 characters.

-Error Handling:

Duplicate username → “Username already exists”.

Invalid email format → “Invalid email format”.

Incorrect password → “Incorrect password”.

Missing field → “This field is required”.

FR-02: Create Quiz

-Description: The system must allow Teachers to create a new quiz with basic information and a list of questions.

-Acceptance Criteria:

The teachers can enter Title, Description, and Time Limit.

The quiz was created successfully.

Quiz is displayed in the list.

-Business Rules:

BR-01: Quiz must contain at least 1 question.

BR-02: Time limit must be greater than 0.

BR-03: Title cannot be left blank.

-Error Handling:

Missing title → “Title is required”.

Time ≤ 0 → “Invalid time limit”.

No questions → “Quiz must contain at least one question”.

FR-03: Manage Questions

-Description: The system allows Teachers to add, edit, or delete questions within a quiz.

-Acceptance Criteria:

The teachers successfully adds/edits/deletes questions.

Display the updated question list.

-Business Rules:

BR-01: Each question must have at least 2 options.

BR-02: There must be at least 1 correct answer.

BR-03: Question content cannot be left blank.

-Error Handling:

Missing content → “Question content is required”.

No correct answer → “At least one correct answer required”.

Less than 2 options → “Minimum 2 options required”.

FR-04: Take Quiz

-Description: The system allows Students to participate in a quiz within the specified time.

-Acceptance Criteria:

The student successfully starts the quiz.

Can select an answer for each question.

The system saves answers temporarily.

-Business Rules:

BR-01: Only 1 answer can be selected per question.

BR-02: Taking the quiz is not allowed after time expires.

-Error Handling:

Time expired → “Time is up”.

No answer selected → Allow skipping or show a warning.

FR-05: Submit Quiz

-Description: The system allows Students to submit the quiz and send answers for grading.

-Acceptance Criteria:

Student successfully submits the quiz.

Display submission confirmation.

-Business Rules:

BR-01: Only 1 submission per attempt. Multiple attempts allowed if configured

BR-02: No editing after submission.

-Error Handling:

Resubmission attempt → “Quiz already submitted”.

No questions answered → Warning: “No answers selected”.

FR-06: Auto Grading

-Description: The system automatically grades the quiz based on correct answers.

-Acceptance Criteria:

Calculate score accurately.

Save results to the system.

-Business Rules:

BR-01: Each correct answer = 1 point (or configurable).

BR-02: Maximum score = total number of questions.

-Error Handling:

Missing attempt data → “Cannot calculate score”.

FR-07: View Result

-Description: The system allows Students can view results depending on visibility configuration.

-Acceptance Criteria:

Display score.

Display correct/incorrect answers.

-Business Rules:

BR-01: Users can only view their own results.

-Error Handling:

Not submitted → “Result not available”.

FR-08: Quiz Management

-Description: The system allows Teachers to manage (view/edit/delete) quizzes.

-Acceptance Criteria:

The teacher successfully edits the quiz.

Quiz is updated.

-Business Rules:

BR-01: Cannot delete a quiz that has existing attempts (optional).

-Error Handling:

Quiz is in use → “Cannot delete active quiz”.

FR-09: Timer System

-Description: The system displays and manages the duration for each quiz.

-Acceptance Criteria:

Display countdown timer.

Automatically submit when time expires.

-Business Rules:

BR-01: Time is calculated immediately upon starting the quiz.

-Error Handling:

Timer error → “Timer error occurred”.

FR-10: User Management

-Description: The system allows Admins to manage users within the system.

-Acceptance Criteria:

Admin adds/edits/deletes users.

Users are assigned the correct permissions.

-Business Rules:

BR-01: Roles include Admin and Student.

BR-02: Cannot delete the main Admin.

-Error Handling:

Deleting main admin → “Cannot delete main admin”.

User does not exist → “User not found”.

FR-11: Role-based Access Control

-Description: The system must authorize users by roles (Student, Teacher, Admin) to control access to functions.

-Acceptance Criteria:

Users only see functions based on their role.

Teacher manages quizzes and question banks.

Admin manages the entire system.

-Business Rules:

BR-01: Roles include Student, Teacher, and Admin.

BR-02: Each user has only 1 role.

-Error Handling:

Unauthorized access → “Access denied”.

FR-12: Change Password

-Description: Users can change their password after logging in.

-Acceptance Criteria:

Enter correct current password.

New password is valid.

-Business Rules:

BR-01: New password ≠ old password.

BR-02: Must comply with policy.

-Error Handling:

Incorrect old password → “Incorrect current password”.

FR-13: Quiz Scheduling & Assignment

-Description: Teacher schedules and assigns quizzes to users.

-Acceptance Criteria:

Select start time & end time.

Assign to student/group.

-Business Rules:

BR-01: Start < End.

BR-02: Must include at least 1 student.

BR-03: Only assigned users can take the quiz.

-Error Handling:

Incorrect time → “Invalid time range”.

Not assigned → “Quiz not available”.

FR-14: Attempt Management

-Description: System manages the number of quiz attempts for students.

-Acceptance Criteria:

Limit the number of attempts.

Save multiple attempts separately.

-Business Rules:

BR-01: Attempt count ≤ max attempts.

BR-02: Each attempt has its own timestamp.

-Error Handling:

Exceeded limit → “Maximum attempts reached”.

FR-15: Auto Save Answers

-Description: System automatically saves answers while the student is taking the quiz.

-Acceptance Criteria:

Save when switching questions.

No data loss on page reload.

-Business Rules:

BR-01: Save on every action.

-Error Handling:

Save error → “Failed to save answer”.

FR-16: Result Visibility Control

-Description: System controls when students can view results.

-Acceptance Criteria:

Display according to configuration.

-Business Rules:

BR-01: Immediate.

BR-02: After end time.

BR-03: Manual release.

-Error Handling:

Not permitted yet → “Result not available yet”.

FR-17: Reporting & Export

-Description: System provides detailed reports.

-Acceptance Criteria:

View attempt list.

Export to CSV.

-Business Rules:

BR-01: Display all attempts.

BR-02: Highlight best score.

-Error Handling:

No data → “No results available”.

FR-18: Quiz & Attempt Deletion

-Description: Manage the deletion of quizzes and attempts.

-Acceptance Criteria:

The teacher deletes the quiz if there are no attempts.

Admin can perform a purge.

-Business Rules:

BR-01: Teacher cannot delete quiz with attempts.

BR-02: Admin has permission to delete everything.

-Error Handling:

Violation → “Cannot delete quiz with attempts”.

FR-19: Provide Answer

-Description: The system must allow the Teacher to provide the correct answer for each question in the question bank.

-Acceptance Criteria:

Teacher can select 1 correct answer when creating or editing a question.

Correct answer is saved and used for grading.

Display correct answer when viewing results (if permitted).

-Business Rules:

Each question has only 1 correct answer.

Answer must be in the options list (A, B, C, D).

-Error Handling:

Correct answer not selected → “Correct answer is required”.

Multiple answers selected → “Only one correct answer allowed”.

FR-20: Edit Answer

-Description: The system must allow the Teacher to edit the answer of a question.

-Acceptance Criteria:

The teacher can change the correct answer.

The system updates data successfully.

-Business Rules:

Editing is not allowed if the question is in a quiz with attempts (optional rule).

-Error Handling:

Question in use → “Cannot modify answer for active quiz”.

Invalid data → “Invalid answer data”.

FR-21: View Notifications

-Description: The system must provide notifications to users about related activities (new quiz, results, system changes).

-Acceptance Criteria:

Users can view a list of notifications.

Notifications display in reverse chronological order.

Can mark as read.

-Business Rules:

Notifications are created upon events (quiz assigned, result published...).

Each user only sees their own notifications.

-Error Handling:

No notifications → “No notifications available”.

Load error → “Failed to load notifications”.

FR-22: User Profile Management

-Description: System allows users to view and update personal information.

-Acceptance Criteria:

Users can view profiles (username, email, role).

Can update information (email, password...).

Changes saved successfully.

-Business Rules:

Email must be in the correct format.

Username cannot be changed (optional).

-Error Handling:

Wrong email → “Invalid email format”.

Save failed → “Update failed”.

II. Non-Functional Requirements (NFR)

NFR Priority Classification (MoSCoW)

Priority	Requirements
MUST HAVE	Performance, Security, Availability, Usability, Reliability, Data Integrity, Deployment, Error Handling, Session Management, Demo Readiness, API Response Consistency, Referential Integrity
SHOULD HAVE	Maintainability, Compatibility, Audit Logging, Data Export, Code Quality, UI Consistency, Privileged Deletion Control
COULD HAVE	Scalability, Backup & Recovery, Accessibility

NFR-01: Performance

-Description: The system must respond quickly to core operations during the demo.

-Criteria: API response ≤ 500ms (≤ 20 concurrent users); Main page load ≤ 3s (4G); Auto-save ≤ 1s; CSV generation ≤ 5s (≤ 1000 records).

-Priority: Must Have

NFR-02: Security

-Description: System must protect user data and control access.

-Criteria: Passwords hashed with bcrypt (cost ≥ 10); Apply RBAC; Session timeout after 30 mins of inactivity.

-Priority: Must Have

NFR-03: Availability

-Description: System must remain stable throughout the demo.

-Criteria: No crash during ≥ 2 hours of continuous demo; System restart time ≤ 2 mins.

-Priority: Must Have

NFR-04: Usability

-Description: System must be easy to use for new users.

-Criteria: Start quiz in ≤ 3 steps; 90% completion rate without guidance.

-Priority: Must Have

NFR-05: Reliability

-Description: The system must ensure no data loss during the quiz.

-Criteria: Auto-save success ≥ 95%; Page reload does not result in data loss.

-Priority: Must Have

NFR-06: Maintainability

-Description: System must be easy to understand and maintain.

-Criteria: README setup guide ≤ 10 mins; Modularized code.

-Priority: Should Have

NFR-09: Data Integrity

-Description: System must ensure consistent data.

-Criteria: No orphan data; Transactions ensure consistency.

-Priority: Must Have

NFR-11: Deployment

-Description: System must be easy to run in the demo environment.

-Criteria: Backend runs via Docker (≤ 5 mins setup); Successful Sybase connection.

-Priority: Must Have

NFR-13: Error Handling (General)

-Description: System must handle errors gracefully.

-Criteria: No UI crashes; 100% of errors have clear messages.

-Priority: Must Have

NFR-15: Session Management

-Description: System manages login sessions.

-Criteria: Timeout after 30 mins; Logout invalidates session.

-Priority: Must Have

NFR-17: Demo Readiness

-Description: System must be fully prepared for a smooth presentation.

-Criteria: Sample data available; Complete demo flow (login → take quiz → view result).

-Priority: Must Have

NFR-19: API Response Consistency

-Description: All APIs must return data in a uniform format.

-Criteria: 100% APIs return standard JSON; Standard HTTP status codes.

-Priority: Must Have

NFR-21: Referential Integrity

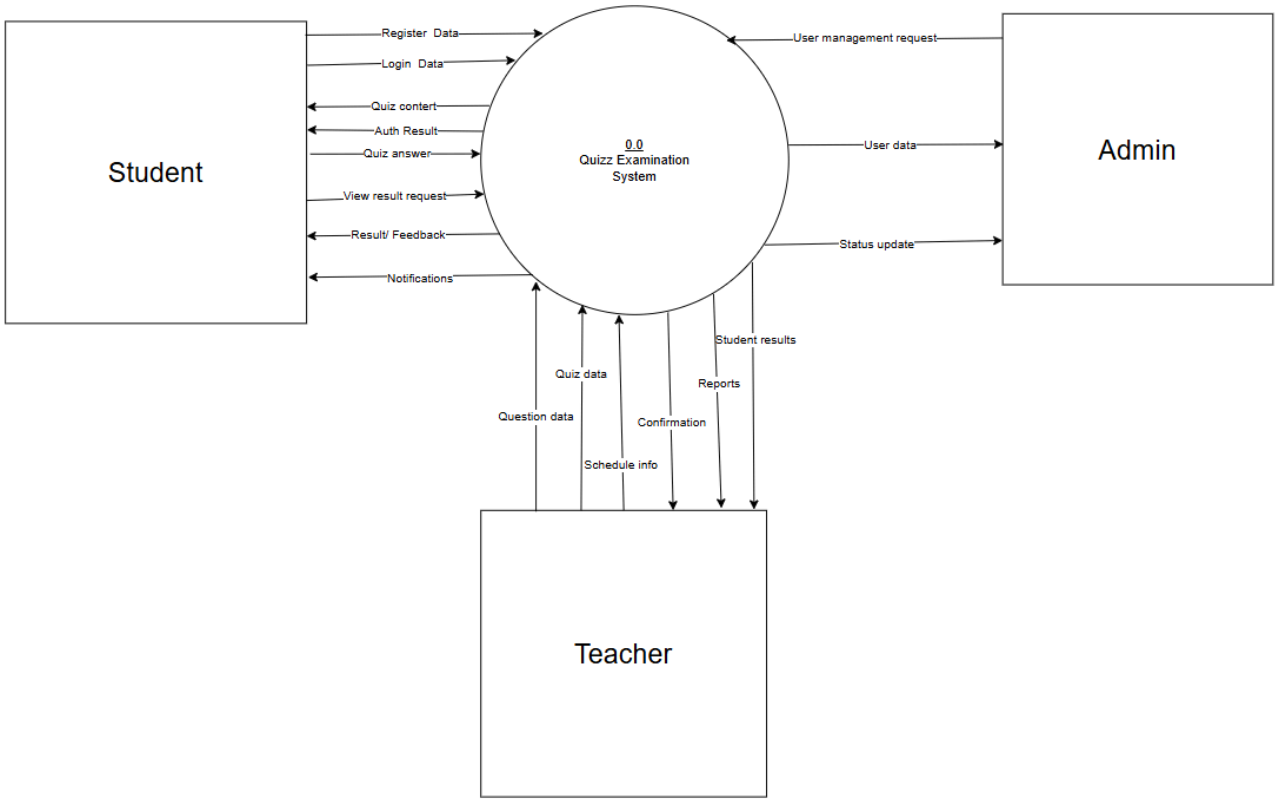
-Description: Ensure no orphan data when performing related deletions.

-Criteria: Deleting attempt removes related answers; No records exist without a relation.

-Priority: Must Have

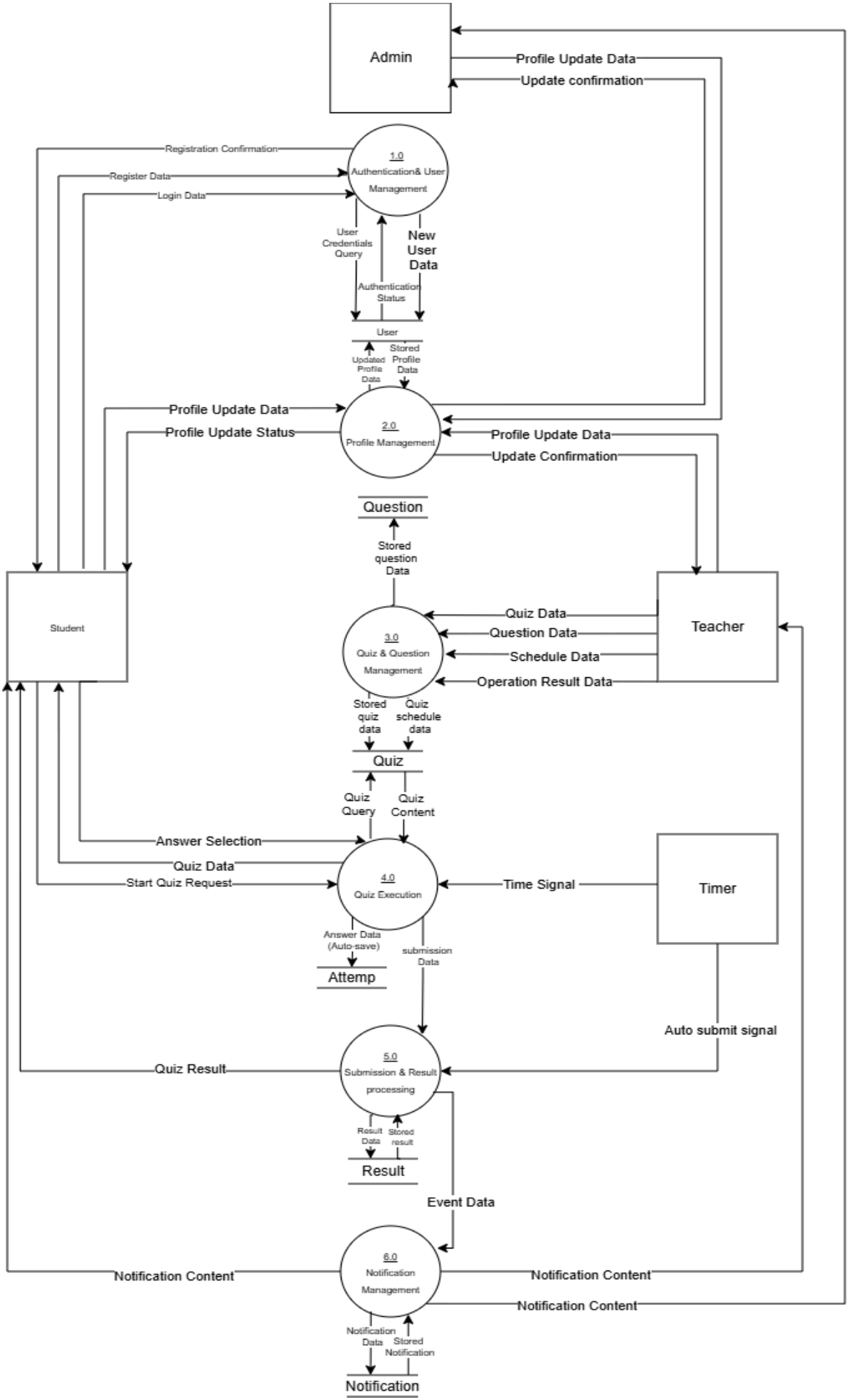
III. Data flow diagram(DFD):

1.DFD level 0



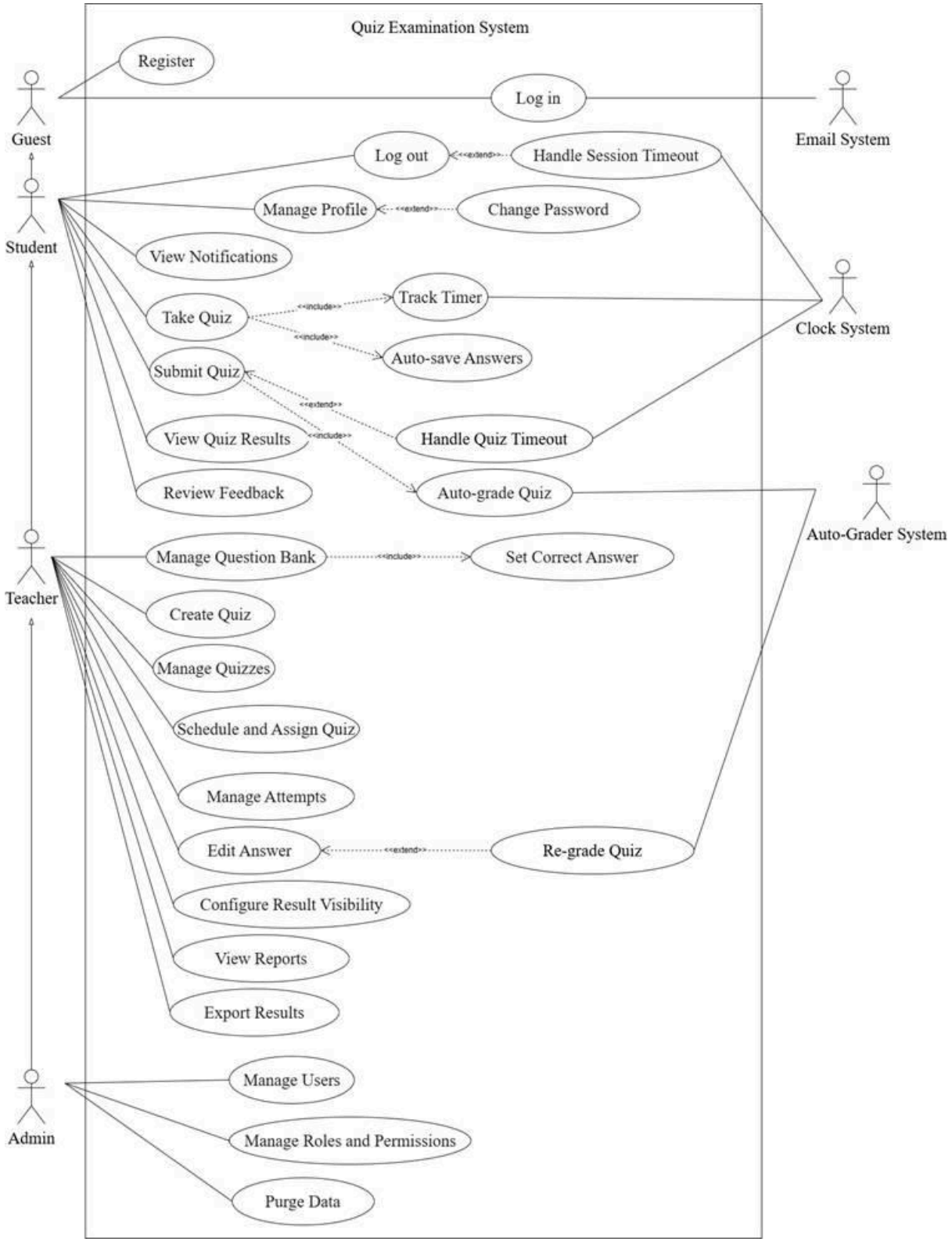
2.DFD level 1

Quiz examinations System	version: 1.1
System specification	Date: 6/4/2026



IV. USE-CASE DIAGRAM

1. Overview Diagram



1.1 List of Actors

No.	Actor	Description
1	Guest	Unauthenticated users who have not yet logged into the system
2	Student	Authenticated users with "Student" privileges and access to quiz-taking features.
3	Teacher	Authenticated users with "Teacher" privileges, responsible for managing questions and quizzes.
4	Admin	System administrators with full control over users, roles, and system maintenance.
5	Email System	An external service used to send verification codes via email.
6	Auto-Grader System	An automated module that calculates scores by matching student answers with the correct ones in the database
7	Clock System	Automated system responsible for monitoring quiz duration to trigger auto-submissions and managing session expiration for security.

1.2 List of Use Cases

No.	Use Case	Description
01	Register	Allows Guests to create a new account as a Student or Teacher.

02	Log In	Authenticates user identity and grants access based on their specific role.
03	Log Out	Ends the current session securely and removes temporary access data.
04	Handle Session Timeout	Automatically logs out users after 30 minutes of inactivity to ensure security.
05	Manage Profile	Updates personal information, profile pictures, and contact details.
06	Change Password	Allows users to update their credentials to maintain account security.
07	View Notifications	Displays system alerts new quizzes, results, or updates in reverse chronological order.
08	Take Quiz	Provides the interface for students to answer questions within a set time limit
09	Track Timer	(Internal Logic) Monitors the time spent during a quiz session in real-time.
10	Auto-save Answers	(Internal Logic) Periodically saves student progress to prevent data loss during a quiz.
11	Submit Quiz	Finalizes the attempt and records the official completion time
12	Handle Quiz Timeout	Triggers an automatic submission when the quiz timer reaches zero.
13	Auto-grade Quiz	(Internal Logic) Automatically evaluates the quiz by matching answers with the key.

14	View Quiz Results	Shows the overall score and basic performance metrics after submission.
15	Review Feedback	Displays detailed correct/incorrect answers along with teacher explanations.
16	Manage Question Bank	Allow teachers to perform CRUD operations on questions and organize by topic.
17	Set Correct Answer	(Internal Logic) Designates the valid response for each question in the bank.
18	Create Quiz	Sets the title, time limit, and maximum attempts for a new assessment.
19	Manage Quizzes	Allows teachers to edit, delete, or hide quizzes that have no attempts.
20	Schedule and Assign Quiz	Sets specific start/end times and assigns students or classes to a quiz.
21	Manage Attempts	Allows teachers to monitor, reset, or delete individual student quiz attempts.
22	Edit Answer	Updates the correct keys for questions after they have been created.
23	Re-grade Quiz	(Internal Logic) Automated process that recalculates scores for all existing student attempts based on updated correct answers to ensure grading accuracy.
23	Configure Result Visibility	Determines when students can see their scores and detailed answers.
24	View Reports	Generates statistical charts and score distributions for an entire class.

25	Export Results	Downloads score data into CSV for external reporting.
26	Manage Users	Allows administrators to lock, unlock, or delete system-wide accounts.
27	Manage Roles and Permissions	Configures access rights for each user group within the system.
28	Purge Data	Permanently removes old logs or redundant records to optimize the database.

2. UC-01: Register

2.1 Summary

This use case describes the process of a Guest creating a new account. The system allows users to register as a Student or a Teacher to access the system with appropriate permissions.

2.2 Flow of Events

2.2.1 Basic Flow

1. Guests access the "Register" screen.
2. The system displays the registration form: Username, Email, and Password.
3. The system provides Role Selection (choose between "Student" and "Teacher").
4. Guests enter personal information and select their desired role.
5. If "Teacher" is selected: The system requests an additional Teacher Validation Code.
6. Guest clicks the "Register" button.

7. The system performs data validation:

- Check for empty fields (Title, Content, or required user fields).
- Verify Email format and uniqueness of Username and Email (BR-01, BR-02).
- Verify password length is at least 8 characters (BR-03).
- If Teacher: Verify the validation code matches the system's configuration.

8. The system performs Password Hashing using bcrypt.
9. The system saves the information into the database with the selected role_id.
10. The system displays "Registration Successful" and redirects to the Login page.

2.2.2 Alternative Flows

- Invalid Information: System displays error messages for duplicates, invalid email format, or weak passwords.
- Wrong Teacher Code: If registering as a Teacher with an incorrect code, the system displays "Invalid Teacher Code" and prevents account creation.
- Missing Information: System displays "This field is required" for any blank mandatory fields.

2.3 Special Requirements

- Security: Passwords must be hashed with bcrypt ($cost \geq 10$) (NFR-02).
- Performance: API response time for registration must be $\leq 500ms$ (NFR-01).
- Data Integrity: Transactions must ensure the account and role are saved simultaneously to avoid "orphan" accounts (NFR-09).

2.4 Pre-conditions

- The user is currently a Guest (unauthenticated).
- The Teacher Code is already configured in the backend system.

2.5 Post-conditions

- A new record is successfully created in the Users table with the correct role.
- The user can now use this account to perform

2.6 Extension Points

- None.

3. UC-02: Log In

3.1 Summary

This use case describes the process where a user (Student, Teacher, or Admin) authenticates their identity to access the system. The system verifies the user's credentials and grants access to specific features based on their assigned role.

3.2 Flow of Events

3.2.1 Basic Flow

1. Guest accesses the "Log in" page.
2. The system displays a login form containing: Username/Email and Password.
3. Guests enter their credentials and click the "Log in" button.
4. The system performs the authentication process:
   - Verifies if the account exists in the database.
   - Uses a password comparison algorithm (matching bcrypt hashes).

5. If the credentials are correct:
   - The system initializes a Session or generates an authentication token (JWT Token).
   - The system identifies the user's Role.
6. The system redirects the user to the corresponding Dashboard:
   - Student: Redirects to the quiz list.
   - Teacher: Redirects to question bank and quiz management.
   - Admin: Redirects to user and system management.

3.2.2 Alternative Flows

- Incorrect Credentials: If the Username/Email does not exist or the password is incorrect, the system displays: "Incorrect username or password".
- Missing Information: If any field is left blank, the system displays: "This field is required".
- Locked Account: If the account has been disabled by an Admin, the system displays: "Your account has been locked".
- Connection Error: If the database connection fails, the system displays: "Server error, please try again later".

3.3 Special Requirements

-Security: Password information must be transmitted over the HTTPS protocol.

- Encryption: Use the bcrypt library to compare the input password with the hashed password in the database.

-Performance: Authentication response time must be $\leq 500ms$ (NFR-01).

-Session Management: After 30 minutes of inactivity, the system must automatically terminate the session (NFR-15).

3.4 Pre-conditions

- The user is currently in an unauthenticated state (Guest).
- The user must have an existing account stored in the Users table.

3.5 Post-conditions

- The session is established, and the system remembers the user's login state.
- The user is granted access to functions according to their role (Role-based Access Control).

3.6 Extension Points

- Forgot Password: Allows users to recover their password via Email if they fail to login multiple times or forget their credentials.

4 UC-03: Log Out

4.1 Summary

This use case describes the process of an authenticated user (Student, Teacher, or Admin) actively terminating their current session. The system revokes access rights and reverts the user to the Guest state to ensure account security.

4.2 Flow of Events

4.2.1 Basic Flow

1. The user clicks the "Log out" button or icon (typically located in the navigation menu or Profile page).
2. The system displays a confirmation dialog: "Are you sure you want to log out?".
3. The user confirms the action.
4. The system executes the logout procedure:
    - Sends a request to the Server to invalidate the current session or token.

- Removes the authentication credentials (JWT Token) or Session data from the browser (Cookies/LocalStorage).
- Clears temporary client-side data regarding the user's Role and personal information.

5. The system redirects the user to the Log in page or the Guest homepage.
6. The system displays a notification: "Logged out successfully".

4.5.1 Alternative Flows

- Cancel Logout: If the user selects "Cancel" in step 3, the system closes the dialog and maintains the current authenticated state.
- Network Error: If the system cannot reach the server to invalidate the session, it prioritizes clearing client-side authentication data to ensure the user cannot continue performing actions on that specific device.

4.6 Special Requirements

- Security: After logging out, the browser's "Back" button must not allow the user to return to protected pages; every internal page must re-verify token validity.
- Synchronization: If the user is currently taking a quiz (UC-08: Take Quiz), the system must provide a warning to submit or ensure progress is saved before logging out to prevent data loss (related to FR-15).
- Performance: The logout process must be highly responsive, with a processing time of $\leq 300ms$.

4.7 Pre-conditions

- The user is currently in an Authenticated state with a specific role (Student, Teacher, or Admin).
- The current Session is still valid.

4.8 Post-conditions

- The session is completely destroyed.

- All subsequent requests to internal functions (Dashboard, Quizzes, Reports) are denied, requiring a new login.
- The user's status reverts to Guest.

4.6 Extension Points

- UC-04: Handle Session Timeout: This use case extends UC-03. It automatically triggers the logout flow if the user remains inactive for 30 minutes, ensuring system security without manual intervention (NFR-15).

5 UC-04: Handle Session Timeout

5.1 Summary

This use case describes the system's automated process of terminating a user session (Student, Teacher, or Admin) after a specific period of inactivity. The system revokes access rights and requires re-authentication to ensure information security.

5.2 Flow of Events

5.2.1 Basic Flow

1. The system initiates an Inactivity Timer immediately after the user's last interaction.
2. The user performs no actions (clicking, typing, or navigating) for a duration of $T \geq 30$ minutes.
3. When the timeout limit is reached, the system executes an automated logout by extending UC-03: Log Out:
   - Sends a request to the Server to invalidate the current authentication token.
   - Clears JWT Tokens or Session Cookies from the browser (LocalStorage/SessionStorage).
   - Removes temporary role and profile data associated with the session.

4. The system redirects the user to the Log in page.
5. The system displays a notification: "Session expired. Please log in again to continue.".

5.6.1 Alternative Flows

- User Re-interaction: If the user performs any action before the limit is reached, the system automatically resets the timer to zero.

In-Progress Quiz (Take Quiz):

- Two minutes before expiration, the system displays a warning dialog: "Your session is about to expire. Do you want to stay logged in?".
- If the student selects "Stay logged in", the system performs a token refresh and resets the timer.
- If there is no response, the system performs a final Auto-save (UC-10) of the current answers before logging out to prevent data loss.

5.7 Special Requirements

- Security: After a timeout, the browser's "Back" button must not allow access to administrative or quiz pages (verified via Middleware).
- Synchronization: Timeout duration must be synchronized between the Client (for UI warnings) and the Server (for API rejection).
- User Experience: Notifications must be clear to prevent user confusion upon redirection.

5.8 Pre-conditions

- The user is in an Authenticated state.
- -The application is active in the user's browser with the Clock System monitoring the session.

5.9 Post-conditions

- All access rights are revoked.
- The old authentication token is completely invalidated on the system.
- The user status reverts to Guest.

5.10 Extension Points

- Extends UC-03: Log Out: This use case triggers the logout procedure automatically.
- UC-02: Log In: The user must re-authenticate to start a new session.

6 UC-05: Manage Profile

6.1 Summary

This use case describes how a user (Student, Teacher, or Admin) views and updates their personal information within the system, including basic details and password changes to enhance security.

6.2 Flow of Events

6.2.1 Basic Flow

1. The user selects "Profile" or "Account Settings" from the navigation bar.
2. The system queries the database and displays current information: Username (Read-only), Email, Full Name, and Role.
3. User modifies the desired fields (e.g., updating to a new Email or Full Name).
4. (Optional) User enters their current password and a new password to perform a password change.
5. The user clicks the "Save Changes" button.
6. The system performs validation checks:
   - Verifies Email format and ensures it is unique among other users (BR-02).

- If changing the password: Validates that the current password is correct and the new password is at least 8 characters long (BR-03).

7. The system updates the new information in the Users table.
8. The system displays a notification: "Profile updated successfully".

6.6.1 Alternative Flows

- Invalid Email Format: The system displays an error: "Invalid email format".
- Email Already Exists: The system displays an error: "Email already exists".
- Incorrect Current Password: The system displays an error: "Incorrect current password" and blocks the password change.
- Invalid New Password: The system displays an error: "New password must be at least 8 characters".
- Update Failed: If a technical error occurs, the system displays: "Update failed, please try again".

6.7 Special Requirements

- Username Constraint: According to business rules, the Username is a unique identifier and cannot be changed after registration (FR-22).
- Security: Changing the password strictly requires the "Current Password" for identity verification.
- Encryption: The new password must be re-hashed using bcrypt before being saved to the database.

6.8 Pre-conditions

- The user has successfully logged into the system.
- The system has a stable connection to the database to retrieve profile information.

6.9 Post-conditions

- The user's personal information is updated in the database.
- If the user changed their password, the current session may require a re-login depending on security configurations.

6.10 Extension Points

- Upload Avatar: Allows users to upload a personal profile picture (typically stored via Supabase Storage).

7 UC-06: Change Password

7.1 Summary

This use case describes the process by which an authenticated user (Student, Teacher, or Admin) updates their password to maintain account security.

7.2 Flow of Events

7.2.1 Basic Flow

1. User accesses the "Change Password" section (within Profile or a dedicated security page).
2. The system displays a form requesting: Current Password, New Password, and Confirm New Password.
3. User enters the required information and clicks "Update Password".
4. The system performs the following validations:
    - Verifies the Current Password matches the one stored in the database.
    - Checks that the New Password complies with the security policy (at least 8 characters).
    - Ensures the New Password is different from the Current Password (BR-01).
    - Confirms that the New Password and Confirm New Password fields match.

5. The system hashes the New Password using bcrypt.
6. The system updates the password record in the Users table.
7. The system displays a success message: "Password changed successfully".

7.6.1 Alternative Flows

- Incorrect Current Password: The system displays an error: "Incorrect current password" and rejects the change.
- Password Mismatch: If the "New" and "Confirm" passwords do not match, the system displays: "Passwords do not match".
- Weak Password: If the new password is less than 8 characters, the system displays: "New password must be at least 8 characters".
- Same as Old Password: If the new password is identical to the old one, the system displays: "New password cannot be the same as the old password".

7.7 Special Requirements

- Security: Password data must be transmitted via HTTPS.
- Encryption: The system must use bcrypt with a cost factor ≥ 10 to hash the new password.
- Session Integrity: Upon a successful password change, the system may invalidate other active sessions to ensure security.

7.8 Pre-conditions

- The user is currently Authenticated and has an active session.

7.9 Post-conditions

- The user's password is encrypted and updated in the database.
- The user must use the new password for all future login attempts.

7.6 Extension Points

- None.

8 UC-07: View Notifications

8.1 Summary

This use case describes how a user (Student, Teacher, or Admin) views a list of system notifications, such as new quiz assignments, published results, or system updates. This feature ensures users stay informed about timely activities.

8.2 Flow of Events

8.2.1 Basic Flow

1. The user clicks the "Notification" icon (typically a bell icon on the navigation bar).
2. The system queries the database to retrieve notifications specific to that user.
3. The system displays the notification list in reverse chronological order (newest first).
4. Each notification displays: Title, a brief content summary, timestamp, and status (Read/Unread).
5. User clicks a specific notification to view details.
6. The system marks that notification as "Read" and updates the database.
7. The system redirects the user to the related page (e.g., clicking a "New Quiz" notification redirects to the Take Quiz page).

8.6.1 Alternative Flows

- No Notifications: If the database returns no records, the system displays: "No notifications available".

- Data Loading Error: If a connection or query error occurs, the system displays: "Failed to load notifications".
- Mark All as Read: The user can select a "Mark all as read" button to update the status of the entire list simultaneously.

8.7 Special Requirements

- Privacy: Users can only view notifications addressed to them specifically or to their assigned Role (Student/Teacher/Admin).
- Real-time (Optional): The system should update the unread notification count immediately without a page reload (using Supabase Real-time).
- Performance: The notification list must load within $\leq 500ms$ to ensure a smooth user experience.

8.8 Pre-conditions

- The user has successfully logged into the system.
- System events have occurred to trigger notifications (e.g., a Teacher assigned a new quiz).

8.9 Post-conditions

- Notification statuses are updated from "Unread" to "Read".
- The unread count on the bell icon decreases accordingly.

8.10 Extension Points

- Push Notifications: Sends alerts directly to the browser or app even when the notification tab is not open.

9 UC-08: Take Quiz

9.1 Summary

This use case describes the process of a Student taking an online quiz. The system provides an interface to display questions, manage the duration, and automatically save the student's selections to prevent data loss.

9.2 Flow of Events

9.2.1 Basic Flow

1. Students select a quiz from their assigned list.
2. The system verifies prerequisites: valid time window and remaining attempt count.
3. The system displays the quiz interface including: Quiz Title, question list, answer options, and a countdown timer.
4. The system activates UC-09: Track Timer: The start time is recorded immediately upon interface display.
5. The student reads the questions and selects an answer (Only 1 answer allowed per question - BR-01).
6. The system triggers UC-10: Auto-save Answers: Whenever a student selects or changes an answer, a draft is saved to the database (NFR-01: save time $\leq$ 1s).
7. Student can navigate between questions or reload the page if necessary.
8. Upon completion, Students click "Submit" to finalize the quiz (Link to UC-11).

9.6.1 Alternative Flows

- Time Expired: When the timer reaches zero, the system notifies "Time is up", locks all inputs, and executes the Auto-submit procedure.
- Network Interruption: If the connection is lost, the system displays a warning: "Failed to save answer" and prompts the user to check their connection.

- Page Reload: The system fetches previously auto-saved answers for display and continues the countdown based on real-time server data (NFR-05).
- Access After Expiration: If a student attempts to access a quiz after the teacher's scheduled end time, the system displays: "Quiz not available".

9.7 Special Requirements

- Reliability: The system must ensure auto-save success $\geq$ 95% of operations (NFR-05).
- Time Synchronization: Quiz duration must be calculated and controlled Server-side to prevent browser-side clock manipulation.
- Performance: The quiz interface must be optimized for fast loading on 4G networks (NFR-01).

9.8 Pre-conditions

- Student is logged into the system.
- The quiz is scheduled and assigned to the student/group (FR-13).
- Current time is within the allowed window (Start < Now < End).
- The student's attempt count is less than the maximum limit (FR-14).

9.9 Post-conditions

- A new Attempt record is initialized and updated continuously.
- Temporary answers are securely stored in the database.
- The system is ready for the automated grading step (UC-13: Auto-grade Quiz).

9.10 Extension Points

- UC-11: Submit Quiz: The end point of the taking process to move to formal submission status.

10 UC-09: Track Timer

10.1 Summary

This internal use case describes how the system automatically monitors and manages the remaining time for a student during a quiz session. It ensures the countdown is accurate and triggers the necessary actions when time expires.

10.2 Flow of Events

10.2.1 Basic Flow

1. The system initiates the countdown timer immediately when the student successfully starts UC-08: Take Quiz.
2. The system retrieves the Time Limit configured for the quiz from the database.
3. The system calculates the Remaining Time by subtracting the elapsed time from the total time limit.
4. The system updates the countdown display on the student's interface every second.
5. When the timer reaches zero (Time = 0), the system automatically triggers UC-12: Handle Quiz Timeout to submit the paper.

10.6.1 Alternative Flows

- Page Reload/Interruption: If the student reloads the page or loses connection, the system re-calculates the remaining time based on the server-side Start Time to prevent time manipulation.
- Timer Synchronization Error: If the client-side clock deviates significantly from the server, the system resynchronizes the timer with the server's time.

10.7 Special Requirements

- Server-Side Control: The authoritative time must be managed on the server to prevent students from modifying the browser's local time.

- Precision: The timer must be accurate to within ± 1 second throughout the entire duration of the quiz.
- Low Latency: Timer updates must not interfere with the performance of the quiz interface (NFR-01).

10.8 Pre-conditions

- The student has successfully started a quiz session in UC-08: Take Quiz.
- The Clock System (System Actor) is functional and synchronized with the server.

10.9 Post-conditions

- The remaining time is updated continuously in the database/session.
- When time expires, the quiz is locked and submitted for grading.

10.10 Extension Points

- UC-12: Handle Quiz Timeout: Triggered automatically when the timer reaches zero.

11 UC-10: Auto-save Answers

11.1 Summary

This internal use case describes the system's automated process of periodically or event-driven saving a student's selected answers during a quiz session. This ensures that progress is preserved even in the event of browser crashes, accidental reloads, or temporary network interruptions.

11.2 Flow of Events

11.2.1 Basic Flow

1. The system monitors the student's actions within the quiz interface of UC-08: Take Quiz.
2. The student performs one of the following triggers:

- Selects or changes an answer for a question.
- Navigates (switches) to a different question.

3. The system captures the current question ID and the selected option.
4. The system asynchronously sends the data to the server to update the Temporary Answers table in the database.
5. The system confirms a successful save within $\le$ 1 second (NFR-01).
6. A subtle "Saved" status indicator may be displayed on the UI to reassure the student.

11.6.1 Alternative Flows

- Network Failure: If the save request fails due to connection issues, the system displays a warning message: "Failed to save answer. Please check your connection.".
- Conflict Resolution: If multiple devices are used simultaneously (not recommended), the system prioritizes the most recent timestamp to maintain data integrity.
- Page Recovery: Upon page reload, the system automatically fetches the latest saved answers from the database to restore the student's progress.

11.7 Special Requirements

- Efficiency: The save operation must be non-blocking and executed in the background to avoid interrupting the student's focus.
- Performance: Auto-save response time must be $\leq 1s$ to prevent interface lag (NFR-01).
- Reliability: The system must achieve a success rate of $\geq 95\%$ for all auto-save operations (NFR-05).

11.4 Pre-conditions

- The student is currently engaged in an active quiz session (UC-08).

- The system has an established database connection (e.g., via Supabase).

11.5 Post-conditions

- The most recent answers are successfully stored as drafts in the database.
- Data is ready to be finalized upon UC-11: Submit Quiz.

11.6 Extension Points

- None.

12 UC-11: Submit Quiz

12.1 Summary

This use case describes the process where a Student finalizes their quiz attempt and sends their answers to the system. Upon submission, the system performs automated grading and records the final results.

12.2 Flow of Events

12.2.1 Basic Flow

1. Student clicks the "Submit" button on the quiz interface.
2. The system displays a confirmation dialog showing a summary (e.g., number of answered questions vs. total).
3. Student confirms the submission.
4. The system executes the following backend processes:
    - Records the official submission time (submit_time).
    - Locks the Attempt status to prevent any further modifications by the student (BR-02).
    - Triggers UC-13: Auto-grade Quiz: Matches selected answers with correct ones in the database.

- Calculates the total score: each correct answer equals 1 point (BR-01, BR-02 of FR-06).

5. The system saves the results (final score, count of correct/incorrect answers) to the database.

6. The system displays a success message and redirects the user to the result page or quiz list.

12.2.2 Alternative Flows

- No Answers Selected: If the student hasn't answered any questions, the system displays a warning: "No answers selected. Are you sure you want to submit?".

- Resubmission Attempt: If a student tries to submit a quiz that is already finalized, the system rejects the request with an error: "Quiz already submitted".

- Grading Error: If answer data is missing during calculation, the system logs an error: "Cannot calculate score".

12.3 Special Requirements

- Atomic Transaction: The submission and grading processes must be executed within a single Atomic Transaction to ensure the score always matches the submitted answers (NFR-09).

- Security: Once submitted, any client-side attempts to modify answers must be rejected by the server.

- Performance: Submission and scoring for a 50-question quiz must be completed in $\leq 1s$ (NFR-01).

12.4 Pre-conditions

- The Student is currently performing UC-08: Take Quiz.

- The quiz Attempt status is "In Progress".

12.5 Post-conditions

- The Attempt status is updated to "Submitted".
- The score is calculated and stored permanently.
- Student access to edit answers for this specific attempt is revoked.

12.6 Extension Points

- UC-12: Handle Quiz Timeout: This use case extends UC-11. When the timer reaches zero (Timer = 0), the system automatically triggers the submission flow starting from Step 4.
- UC-14: View Quiz Results: Allows students to view their score immediately after a successful submission (depending on teacher configuration).

12.7 Included Use Cases

- UC-13: Auto-grade Quiz: This process is included as an essential step of the submission workflow to provide immediate feedback.

13 UC-12: Handle Quiz Timeout

13.1 Summary

This internal use case describes the system's automated action when the quiz timer reaches zero. The system forcefully terminates the "Take Quiz" session and initiates the submission process to ensure no further answers can be modified.

13.2 Flow of Events

13.2.1 Basic Flow

1. The Clock System (System Actor) monitors the remaining time in real-time during UC-08: Take Quiz.
2. The countdown reaches zero (Timer = 0).

3. The system immediately locks the user interface, preventing any further interaction with the questions or answers.

4. The system displays a brief notification to the student: "Time is up! Your quiz is being submitted automatically.".

5. The system triggers the automated submission flow by extending UC-11: Submit Quiz (starting from the backend processing steps).

6. The system captures all currently saved answers from UC-10: Auto-save Answers for final grading.

13.2.2 Alternative Flows

- Network Interruption at Timeout: If the student's device is offline when the timer hits zero, the system will attempt to sync and submit as soon as the connection is restored, using the server-side recorded expiration time as the official limit.

- Pre-submission: If the student manually completes UC-11 before the timeout, this use case is not triggered.

13.3 Special Requirements

- Authoritative Timing: The timeout trigger must be controlled by the server-side clock to prevent client-side manipulation.

- Reliability: The transition from "Take Quiz" to "Submit Quiz" must be seamless and fail-safe to prevent data loss (NFR-05).

- UI Lock: The locking mechanism must be robust enough to stop all input methods (keyboard, mouse, touch).

13.4 Pre-conditions

- The student is currently in the middle of UC-08: Take Quiz.

- The Clock System is actively tracking the session.

13.5 Post-conditions

- The quiz attempt is moved to the "Submitted" state.

- The student's access to the quiz content is revoked.
- Results are processed and ready for viewing (if permitted).

13.6 Extension Points

- Extends UC-11: Submit Quiz: This use case provides the automated trigger for the standard submission procedure.

14 UC-13: Auto-grade Quiz

14.1 Summary

This internal use case describes the automated process performed by the Auto-Grader System to calculate a student's score immediately after a quiz is submitted. It ensures high accuracy and provides instant results for objective assessments.

14.2 Flow of Events

14.2.1 Basic Flow

1. This use case is included and triggered automatically by UC-11: Submit Quiz.
2. The system retrieves the student's final answers (saved in UC-10/UC-11) and the corresponding correct answers from the Question Bank (FR-19).
3. The system compares each student answer with the stored correct answer key.
4. The system calculates the score based on the following business rules:
   - Each correct answer = 1 point (unless configured otherwise) (BR-01 of FR-06).
   - Total score = sum of all correct answers (BR-02 of FR-06).
5. The system records the count of Correct, Incorrect, and Unanswered questions.

6. The system saves the final calculated score and performance metrics into the Results table.

14.2.2 Alternative Flows

- Missing Attempt Data: If the system cannot find the student's answer record, it logs an error: "Cannot calculate score" and notifies the Admin.
- Invalid Answer Key: If a question in the quiz lacks a designated correct answer, the system flags the attempt for Teacher review.

14.3 Special Requirements

- Accuracy: The grading logic must be 100% accurate based on the provided answer key.
- Performance: Scoring for a standard quiz (e.g., 50 questions) must be completed in ≤ 500 to ensure immediate feedback (NFR-01).
- Data Integrity: The grading process must be part of the submission Transaction to prevent score-data mismatch (NFR-09).

14.4 Pre-conditions

- UC-11: Submit Quiz has been initiated and the answers are finalized.-
- The Auto-Grader System is functional and has access to the correct answer keys.

14.5 Post-conditions

- The student's score is successfully calculated and persisted in the database.
- The quiz attempt is ready for UC-14: View Quiz Results.

14.6 Include Relationship

- Included by UC-11: Submit Quiz: This process is an essential component of the quiz submission workflow.

15 UC-14: View Quiz Results

15.1 Summary

This use case describes how a Student views their scores and general performance summary after successfully submitting a quiz. The system displays specific information based on the teacher's visibility configuration.

15.2 Flow of Events

15.2.1 Basic Flow

1. Student accesses the "My Attempts" list or clicks the "View Result" button immediately after submission.
2. The system queries the database to retrieve the student's specific Attempt record.
3. The system verifies Visibility Permissions (Logic from FR-16):
   - Checks if the quiz is configured for immediate result display.
4. If conditions are met, the system displays the result dashboard including:
   - Total Score achieved / Maximum score.
   - Time taken (Start time - End time).
   - Number of Correct and Incorrect answers.
5. Student reviews their personal performance on the screen.

15.2.2 Alternative Flows

- Release Time Not Reached: If configured as "View after end time" (FR-16), the system displays: "Result will be available after [End Time]".
- Manual Release Pending: If the quiz is in "Manual release" mode and not yet approved, the system displays: "Result not available yet".

- Unauthorized Access: If a student attempts to view another user's results by altering the URL, the system verifies ownership and displays: "Access denied".
- Unsubmitted Attempt: If the attempt is still "In Progress", the system prompts the student to submit before viewing the score.

15.3 Special Requirements

- Privacy: Strictly adheres to Data Isolation; students can only view their own results (FR-07).
- Performance: Retrieval and calculation of results must be completed within $\leq 500ms$ (NFR-01).
- Accuracy: Scores must perfectly match the data processed by the Auto-Grader System.

15.4 Pre-condition

- Student is logged into the system.
- The quiz attempt is recorded as "Submitted" in the database.

15.5 Post-condition

- No data is changed (Read-only operation).
- The student is informed of their academic performance for the respective quiz.

15.6 Extension Point

- UC-15: Review Feedback: If permitted by configuration, students can click to view detailed correct/incorrect answers and explanations.

16 UC-15: Review Feedback

16.1 Summary

This use case describes how a Student reviews each question in a submitted quiz. The system displays the student's selected answer, compares it with the correct key, and provides teacher feedback (explanations) to help the student understand their performance.

16.2 Flow of Events

16.2.1 Basic Flow

1. Student selects a specific attempt to review from the UC-14: View Quiz Results interface.
2. The system queries the database to retrieve the quiz questions, student's selected answers, the correct answer keys, and corresponding explanations.
3. The system displays the "Review Mode" interface:
   - Each question shows the content and all available options.
   - Clearly marks the student's selected answer.
   - Uses visual cues: Green (✅) for correct answers and Red (❌) for incorrect ones.
   - Displays the Correct Answer if the student answered incorrectly.
   - Displays the Feedback (Explanation) provided by the teacher under each question.
4. Student browses through the list of questions to self-assess.

16.2.2 Alternative Flows

- Feedback Hidden: If the teacher has configured the quiz to restrict detailed review (Logic from FR-16), the system only displays the questions and student's choices without showing correct keys or explanations.
- Data Loading Error: The system displays a notification: "Failed to load feedback details" in case of network or database connection issues.

- No Explanation Provided: If the teacher did not enter an explanation during question creation, the system only displays the correct key and omits the feedback section.

16.3 Special Requirements

- Usability: The interface must be visually intuitive, clearly distinguishing between user choices and correct keys to avoid confusion.
- Security: The system verifies attempt ownership to ensure students cannot view feedback for other users' quizzes by manipulating URLs (Data Isolation).
- Performance: The time to load and display the full list of questions with feedback must be $\leq 1s$ for a standard quiz (NFR-01).

16.4 Pre-conditions

- Student has successfully logged into the system.
- The quiz attempt status is "Submitted".
- Quiz configuration (FR-16) permits students to view detailed answers and feedback.

16.5 Post-conditions

- The system data remains unchanged (Read-only operation).
- The student completes their review of mistakes and explanations.

16.6 Extension Points

- None.

17 UC-16: Manage Question Bank

17.1 Summary

This use case describes how a Teacher manages the repository of questions in the system, including: Adding, Editing, and Deleting questions. This question bank serves as the primary data source for constructing assessments in UC-18: Create Quiz.

17.2 Flow of Events

17.2.1 Basic Flow

1. Teacher accesses the "Manage Question Bank" feature from the navigation menu.
2. The system displays a list of existing questions with filters for subject and difficulty level.
3. Teacher selects one of the following actions:

- Add New Question:

a. The system displays an input form: Question content, options, difficulty, and subject.

b. Teacher enters the content and at least 2 options (BR-01 of FR-03).

c. Included UC-17 (Set Correct Answer): Teacher designates at least one option as the correct answer (BR-02 of FR-03).

d. Teacher clicks "Save".

- Edit Question:

a. Teacher selects a question to modify.

b. Teacher updates the content or answers and clicks "Update".

- Delete Question:

a. Teacher selects a question and confirms the deletion request.

4. The system validates the data (content cannot be blank, correct answer must be set).
5. The system updates the Questions and Answers tables in the database.
6. The system displays a success notification.

17.2.2 Alternative Flows

- Question In Use: If the Teacher attempts to delete a question that is currently part of an active quiz with student attempts, the system displays: "Cannot delete active quiz" (BR-01 of FR-08).
- Missing Data: If required fields are blank or no correct answer is selected, the system displays: "Question content is required" or "At least one correct answer required".
- Query Error: If the database connection fails, the system displays: "Unable to save question. Please try again later".

17.3 Special Requirements

- Consistency: When editing, the system should ideally version questions to ensure past student results remain accurate and unaffected by changes.
- Multimedia Support: Supports inserting images or technical formatting using Markdown or LaTeX for question content.
- Performance: Loading the paginated question bank list must be completed in $\leq 1s$ (NFR-01).

17.4 Pre-conditions

- The user is logged in with Teacher privileges.
- The database is ready for CRUD (Create, Read, Update, Delete) operations.

17.5 Post-conditions

- The database is updated with new or modified question records.
- New questions are available for selection in UC-18: Create Quiz.

17.6 Extension Points

-Import/Export: Allows Teachers to bulk upload questions from CSV files.

18 UC-17: Set Correct Answer

18.1 Summary

This internal use case describes the process where a Teacher designates the valid response for a specific question in the question bank. This correct answer is mandatory for the Auto-Grader System to calculate scores accurately.

18.2 Flow of Events

18.2.1 Basic Flow

1. This use case is included and triggered during the "Add" or "Edit" question process in UC-16: Manage Question Bank.
2. The system displays the list of provided answer options (e.g., A, B, C, D).
3. Teacher selects exactly one option to be the correct answer.
4. The system highlights the selected option to confirm the choice.
5. The system associates the "correct" flag with the chosen answer ID in the database.
6. Upon saving the question in UC-16, the system validates that a correct answer has been assigned.

18.2.2 Alternative Flows

- No Correct Answer Selected: If the Teacher attempts to save without selecting an answer, the system displays: "Correct answer is required".
- Attempting Multiple Correct Answers: If the Teacher tries to select more than one option (in a single-choice setup), the system displays: "Only one correct answer allowed".
- Editing Answer in Active Quiz: If the Teacher tries to change the correct answer for a question already used in a quiz with existing attempts, the system may issue a warning or trigger UC-23: Re-grade Quiz.

18.3 Special Requirements

- Data Integrity: The correct answer ID must correspond to an existing option in the Answers table to prevent orphan records.
- Logic Consistency: If a question is edited, the system must ensure the new correct answer is used for all future grading.

18.4 Pre-conditions

- The Teacher is currently adding or editing a question in UC-16.
- At least two answer options have been entered for the question.

18.5 Post-conditions

- The correct answer is successfully mapped to the question.
- The question is validated and ready for use in any quiz assessment.

18.6 Included Relationship

- Included by UC-16: Manage Question Bank: Setting the answer is a mandatory step in the question creation workflow.

19 UC-18: Create Quiz

19.1 Summary

This use case describes the process by which a Teacher establishes a new assessment by configuring its basic information (title, description, time limit, maximum attempts) and selecting specific questions from the Question Bank created in UC-16.

19.2 Flow of Events

19.2.1 Basic Flow

1. Teacher selects the "Create New Quiz" function from the management dashboard.
2. The system displays a setup form for basic information:
    - Title: The name of the quiz.

- Description: A brief summary of the quiz content.
- Time Limit: Duration allowed for the quiz in minutes.
- Max Attempts: Maximum number of times a student can take the quiz.

3. Teacher enters the configuration details.
4. The system retrieves and displays the question list from the Question Bank (UC-16).
5. Teacher selects the questions to include in the quiz by marking the checkboxes.
6. Teacher clicks the "Save/Create" button.
7. The system performs validation checks:
   - Ensures the Title is not blank (BR-03 of FR-02).
   - Ensures at least one question is selected (BR-01 of FR-02).
   - Verifies that the Time Limit is greater than zero (BR-02 of FR-02).
8. The system saves the quiz into the Quizzes table and establishes links in the QuizQuestions junction table.
9. The system displays a notification: "Quiz created successfully".

19.2.2 Alternative Flows

- No Questions Selected: If the Teacher saves without selecting any questions, the system displays an error: "Quiz must contain at least one question".
- Missing Required Information: If mandatory fields are empty, the system displays: "Title is required".
- Cancel Operation: If the Teacher selects "Cancel," the system returns to the quiz list without saving any data.

19.3 Special Requirements

- Flexibility: Allow Teachers to search and filter questions by category/subject while selecting questions to speed up the process.
- Score Calculation: The system automatically calculates the total quiz score based on the number of selected questions (e.g., 1 point per correct answer).
- Performance: Saving a quiz with 50–100 questions must be completed within ≤ 1 (NFR-01).

19.4 Pre-conditions

- The user is logged in with the Teacher role.
- Questions are already available in the system (UC-16: Manage Question Bank).

19.5 Post-conditions

- A new quiz record is created in the database.
- The quiz is ready to be assigned in UC-20: Schedule and Assign Quiz.

19.6 Extension Points

- Randomize Questions: An option allowing the system to shuffle the question order for each student to prevent cheating.

20 UC-19: Manage Quizzes

20.1 Summary

This use case describes how a Teacher manages existing quizzes in the system. Key operations include viewing the quiz list, updating information, toggling visibility status (Active/Hidden), and deleting quizzes.

20.2 Flow of Events

20.2.1 Basic Flow

1. Teacher accesses the "Manage Quizzes" section from the administrative dashboard.

2. The system queries and displays a list of quizzes created by that specific teacher.

3. The list includes: Title, question count, status (Active/Draft/Hidden), and the total number of student attempts.

4. Teacher selects one of the following actions:

   - View Detail: The system displays the full configuration and question list of the quiz.

   - Update (Edit):

     a. The system checks if any students have already attempted the quiz.

     b. If no attempts exist, the system opens the form to edit the Title, Time Limit, and Question List.

     c. Teacher clicks "Update" to save changes.

   - Toggle Status: Teacher switches between "Active" (visible to students) and "Draft/Hidden" (invisible).

   Delete Quiz: Teacher selects delete and confirms the request in a warning dialog.

5. The system validates the requests and updates the Quizzes table in the database.

6. The system displays a success notification.

20.2.2 Alternative Flows

- Quiz Has Existing Attempts: If a Teacher tries to edit content or delete a quiz that has already been submitted by students, the system displays an error: "Cannot delete active quiz" or "Cannot modify answer for active quiz".

- Delete Confirmation: If the Teacher chooses to delete, the system always prompts for final confirmation: "Are you sure? This action cannot be undone.".
- Query Error: If the connection to Supabase or the database fails, the system displays: "Unable to update quiz status. Please try again.".

20.3 Special Requirements

- Data Integrity: To maintain historical reporting, quizzes with existing results should not be physically deleted (Soft Delete).
- Data Isolation: Teachers are only permitted to manage quizzes that they have personally created.
- Performance: The quiz list and attempt statistics must load in $\leq 1s$.

20.4 Pre-conditions

- The user is logged in with the Teacher role.
- At least one quiz exists in the database.

20.5 Post-conditions

- The quiz information is updated to the latest version.
- The quiz's visibility to students changes accordingly based on the selected status.

20.6 Extension Points

- Clone Quiz: Allows a Teacher to duplicate an existing quiz into a new one for quick editing without affecting historical data.

21 UC-20: Schedule and Assign Quiz

21.1 Summary

This use case describes how a Teacher sets time constraints (Start Time and End Time) and specifies the target audience (individual students or specific classes)

authorized to participate in a quiz. This function ensures that assessments are conducted according to the scheduled curriculum.

21.2 Flow of Events

21.2.1 Basic Flow

1. The teacher accesses the "Schedule & Assign" feature from the quiz management dashboard.
2. The system displays a list of existing quizzes. The teacher selects a quiz to be scheduled.
3. The system displays a scheduling form including:
    - Start Time: Date and time when the quiz becomes available.
    - End Time: Date and time when the quiz expires.
4. Teacher selects the target audience for the assignment:
    - Assign to all students.
    - Assign to one or more specific classes.
    - Assign to specific individual students.
5. The teacher clicks the "Confirm Schedule" button.
6. The system performs validation checks:
    - Ensures the end time is after the start time (End Time > Start Time) (BR-01 of FR-13).
    - Ensures at least one target (student or class) is selected (BR-02 of FR-13).
7. The system saves the schedule information and creates authorization records in the database.
8. The system automatically triggers UC-07: View Notifications: Sends alerts to assigned students about the new quiz.

21.2.2 Alternative Flows

- Time Logic Error: If the Teacher selects an end time earlier than the start time, the system displays: "End time must be after start time".
- Start Time in the Past: If the start time is earlier than the current system time, the system issues a warning for the Teacher to re-confirm.
- No Target Selected: If the Teacher saves without selecting any classes or students, the system displays: "Please select at least one class or student".

21.3 Special Requirements

- Time Synchronization: All time-check logic must be executed Server-side to prevent students from cheating by modifying their local computer clocks.
- Automation: The system must automatically update the quiz status to "Open" at the start time and "Closed" at the end time without manual intervention.
- Performance: Assigning quizzes to large groups (e.g., 1,000+ students) must be processed in the background to prevent interface freezing.

21.4 Pre-conditions

- The user is logged in with the Teacher role.
- The quiz has been successfully created in UC-18: Create Quiz.

21.5 Post-conditions

- The quiz is scheduled and will appear in the students' lists at the designated time.
- Notifications are successfully dispatched to the relevant students.

21.6 Extension Points

- None.

22 UC-21: Manage Attempts

22.1 Summary

This use case describes how a Teacher manages student quiz attempts for a specific assessment. Teachers can monitor progress, view detailed results, or delete/reset attempts in cases where students encounter technical issues or requires a second chance.

22.2 Flow of Events

22.2.1 Basic Flow

1. The teacher selects a specific quiz from the quiz management list.
2. The system displays a list of Attempts associated with that quiz, including: Student Name, Start Time, Submission Time, Status (In Progress/Submitted), and Score.
3. Teacher can perform the following actions:
    - View Detail: The system displays the student's specific answers for that attempt (Redirects to UC-15: Review Feedback under Teacher view permissions).
    - Delete Attempt: Teacher selects one or more attempts and clicks "Delete".
    - Reset Attempt: Teacher selects a failed or interrupted attempt and clicks "Reset". The system clears the previous results and allows the student to restart without consuming their maximum attempt limit.
4. The system requests confirmation before executing delete or reset operations.
5. The system updates the Attempts table and related data in the database.
6. The system displays a notification: "Operation successful".

22.2.2 Alternative Flows

- No Attempts Found: If the quiz is new or no students have started yet, the system displays: "No attempts found for this quiz".

- Locked During Processing: If the system is currently performing UC-13: Auto-grade Quiz for an attempt, the Teacher is restricted from deleting it until grading is complete.
- Search/Filter: Teachers can filter the attempt list by student name or status ("Submitted" vs. "In Progress") for easier management.

22.3 Special Requirements

- Audit Logs: Every action taken by a teacher to delete or reset an attempt must be logged by the system to prevent disputes and maintain transparency.
- Report Impact: When an attempt is deleted, the system must automatically recalculate score distributions and statistics for UC-24: View Reports.
- Performance: Loading the attempt list for a large class (e.g., 100–200 students) must be completed in $\le 1s$ (NFR-01).

22.4 Pre-conditions

- The user is logged in with the Teacher role.
- The quiz has been assigned, and at least one student has started the attempt.

22.5 Post-conditions

- The attempt list is updated (records are removed or states are reset).
- Student access to the quiz is restored if the "Reset" command was executed.

22.6 Extension Points

- Manual Grade Override: Allows the teacher to manually adjust a score if a question contains errors or requires subjective review (linked to UC-22).

23 UC-22: Edit Answer

23.1 Summary

This use case describes how a Teacher updates or changes the correct answer and option text for a question within the question bank. This is typically used to correct pedagogical errors or adjust answers based on student feedback.

23.2 Flow of Events

23.2.1 Basic Flow

1. The teacher accesses the question list within UC-16: Manage Question Bank.
2. The system displays the list of questions. The teacher selects a specific question to edit its answer.
3. The system displays the question details and current answer options.
4. Teacher performs the modifications:
   - Selects a different option to be the correct answer (FR-20).
   - Or edit the text content of the existing options.
5. The teacher clicks the "Update Answer" button.
6. The system performs validation checks:
   - Ensures exactly one option is designated as the correct answer (BR-01 of FR-19).
   - Ensures no answer fields are left blank.
7. The system updates the new information in the Answers table in the database.
8. The system displays a success notification: "Answer updated successfully".

23.2.2 Alternative Flows

- Attempts Exist (Critical Warning): If the question is part of a quiz that has already been submitted by students (status = 'Submitted'), the system issues a warning: "This question has already been answered by students.

Editing the answer may affect their current scores. Do you want to proceed?".

- Cancel Operation: If the Teacher selects "Cancel," the system returns to the previous screen without saving changes.
- System Error: If the data cannot be overridden, the system displays: "Update failed. Please check your connection.".

23.3 Special Requirements

- Logic Constraint: If the system is strictly configured, teachers may only edit answers before a quiz starts. If attempts exist, editing may require Admin approval or trigger a re-grading process.
- Audit Log: The system must record: Who edited it, When, and the Old Answer vs. New Answer to resolve potential student disputes.
- Performance: Updating a single question's answer must occur immediately ($\leq 300ms$).

23.4 Pre-conditions

- The user is logged in with the Teacher role.
- The target question already exists in the database.

23.5 Post-conditions

- The correct answer is updated in the database.
- All future quizzes using this question will apply the new correct answer key.

23.6 Extension Points

- Recalculate Scores (Optional): Triggers a process to automatically update the scores of all students who have already completed this question based on the new answer key.

24 UC-23: Re-grade Quiz

24.1 Summary

This use case describes the automated process of re-calculating student scores for all submitted attempts after a teacher has corrected a question's answer key in UC-22: Edit Answer. This ensures that academic records reflect the most accurate data without requiring students to retake the assessment.

24.2 Flow of Events

24.2.1 Basic Flow

1. This use case is triggered as an extension of UC-22: Edit Answer when the Teacher confirms the answer update and selects the "Re-grade all attempts" option.
2. The system identifies all quiz attempts with a status of "Submitted" that contain the modified question.
3. The system initiates the Auto-Grader System (UC-13) in batch mode to re-evaluate only the affected question for each identified attempt.
4. The system updates the Total Score and performance metrics in the Results table for each student.
5. The system records the re-grading event in the Audit Log, noting the score delta (the difference between the old and new scores).
6. The system displays a summary report to the Teacher: "Re-grading complete. [X] attempts updated."

24.2.2 Alternative Flows

- No Attempts Affected: If the quiz has not been taken yet, the system skips the re-grading process and notifies the Teacher: "No student attempts affected. Re-grading skipped."
- Massive Data Batching: If the quiz has a very large number of attempts (e.g., >1000), the system processes the re-grading as a background task and notifies the Teacher via UC-07: View Notifications once complete.
- Process Interruption: If a database error occurs, the system rolls back the current transaction to ensure no partial or inconsistent scoring occurs.

24.3 Special Requirements

- Data Integrity: The re-grading must be performed within a Database Transaction to ensure that all affected attempts are updated consistently or none at all.
- Audit Trail: The system must preserve a record of the original score before the re-grading for transparency and dispute resolution.
- Performance: Re-grading for a standard class size (e.g., 50–100 students) must be completed in $\leq 2s$.

24.4 Pre-conditions

- UC-22: Edit Answer has been successfully executed for a question that is already part of a submitted quiz.
- The system has a stable connection to the database.

24.5 Post-conditions

- All historical scores for the affected quiz are updated according to the new answer key.
- The updated results are immediately available for students to view in UC-14: View Quiz Results.

24.6 Extension Relationship

- Extends UC-22: Edit Answer: This use case is an optional follow-up action to correcting a question's answer.

25 UC-24: Configure Result Visibility

25.1 Summary

This use case describes how a teacher establishes rules for displaying quiz results to students. The teacher decides when students can view their scores and whether they are allowed to see detailed correct/incorrect answers or explanations.

25.2 Flow of Events

25.2.1 Basic Flow

1. The teacher accesses the "Manage Quizzes" section and selects a specific quiz.
2. The system displays Visibility Settings:

Display Mode:

- Show total score only.
- Show score with correct/incorrect indicators.
- Show score, answers, and detailed explanations (UC-15: Review Feedback).

Release Timing:

- Immediately: Right after the student submits the quiz.
- After Quiz Ends: Only after the quiz's End Time has passed (FR-16).
- Manually: Only when the teacher toggles the release manually.

3. The teacher selects the parameters suitable for the assessment goals.
4. The teacher clicks "Apply Settings".
5. The system validates the configurations and updates the Quizzes table in the database.
6. The system displays a notification: "Visibility settings updated".

25.2.2 Alternative Flows

- Time Logic Error: If "After Quiz Ends" is selected but no End Time is set for the quiz, the system prompts the teacher to set a schedule first in UC-20.
- Cancel Operation: If the teacher exits without saving, the system retains the previous configuration.

- Database Connection Error: If the connection to Supabase fails, the system displays: "Unable to save configuration.".

25.3 Special Requirements

- Immediacy: Changes to visibility settings must take effect instantly for students accessing UC-14: View Quiz Results.
- Security: To prevent cheating, the system must not send correct answer data to the student's browser if the configuration is set to "Hidden" (Backend-level filtering).
- UI/UX: Provide intuitive controls like Toggle switches or Dropdowns for quick teacher interaction.

25.4 Pre-conditions

- The user is logged in as a Teacher.
- The quiz record exists in the database.

25.5 Post-conditions

- Quiz visibility configurations are updated.
- The Student interface updates accordingly (the "View Results" or "Review" buttons appear or disappear based on these rules).

25.6 Extension Points

- None.

26 UC-25: View Reports

26.1 Summary

This use case describes how a Teacher or Admin views comprehensive statistical reports regarding the results of a specific quiz. The system aggregates data from all submitted attempts to provide key metrics such as average scores, pass/fail rates, and score distributions.

26.2 Flow of Events

26.2.1 Basic Flow

1. The teacher selects a specific quiz from the quiz management list and clicks the "View Reports" section.
2. The system queries all records from the Attempts table related to that quiz.
3. The system calculates statistical metrics:
   - Total Attempts: Number of students who submitted the quiz.
   - Average Score: The mean score of all attempts.
   - Highest and Lowest Scores.
   - Pass/Fail Rate: Based on a pre-defined passing threshold (Logic from FR-17).
4. The system displays the report using visual components:
   - Score Distribution (Histogram): Shows the number of students within specific score ranges.
   - Pass/Fail (Pie Chart): Displays the percentage of students who passed vs. failed.
   - Hardest Questions: A list of questions with the highest incorrect response rates.
5. Teachers can filter the report by specific classes or timeframes.

26.2.2 Alternative Flows

- No Data Available: If no students have submitted the quiz yet, the system displays: "No data available to generate report".
- Data Overload (Timeout): If the volume of records is extremely high, causing a delay, the system displays: "Data is being processed, please refresh in a moment".

- Exit Report: If the Teacher leaves the page, the system returns to the management dashboard.

26.3 Special Requirements

- Visualization: The system should utilize charting libraries (such as Chart.js or Recharts) to present data visually rather than just raw tables.
- Accuracy: Statistical calculations must be 100% accurate based on the actual data in the database.
- Update Policy: The report should have a "Refresh" option to pull the latest data if students are currently submitting quizzes.

26.4 Pre-conditions

- The user is logged in as a Teacher or Admin.
- The quiz exists and has at least one "Submitted" attempt in the database.

26.5 Post-conditions

- The system data remains unchanged (Read-only operation).
- The teacher gains insights into the student's learning progress and the quality of the assessment.

26.6 Extension Points

- UC-26: Export Results: From the report interface, teachers can choose to export the data to CSV for archival or submission to the institution.

27 UC-26: Export Results

27.1 Summary

This use case describes how a Teacher or Admin exports student score data from the system into a CSV (Comma-Separated Values) file. This serves the purpose of offline storage, institutional reporting, or data migration into other academic management systems.

27.2 Flow of Events

27.2.1 Basic Flow

1. The teacher accesses the report interface for a specific quiz via UC-25: View Reports.
2. The system displays the "Export Results" functional button.
3. The teacher clicks the Export button.
4. (Optional) The system allows the teacher to filter the data scope (e.g., by specific class) before confirming.
5. Teacher confirms the export request.
6. The system performs a database query and aggregates the following information: Student ID, Full Name, Class, Final Score, and Completion Time.
7. The system generates a CSV file and triggers the Download process on the user's browser.
8. The system displays a notification: "Data exported successfully".

27.2.2 Alternative Flows

- No Data to Export: If the quiz has no recorded attempts, the system disables the Export button or displays: "No results found to export".
- File Generation Error: If the data volume is too large or a server error occurs, the system displays: "Failed to generate file. Please try again later".
- Cancel Operation: The user cancels the request, and the system returns to the report screen.

27.3 Special Requirements

- Encoding (UTF-8): The exported CSV file must support UTF-8 with BOM encoding to ensure Vietnamese characters are displayed correctly when opened in spreadsheet software like Excel.

- Data Integrity: Data in the CSV file must match 100% with the records stored in the database.
- Security: Only authorized users (Teacher/Admin) for that specific quiz are permitted to use the export function (Data Isolation).

27.4 Pre-conditions

- The user is logged in as a Teacher or Admin.
- At least one quiz result (Attempt) has been recorded in the database.

27.5 Post-conditions

- A CSV file containing the score list is successfully saved on the user's local device.
- Database state remains unchanged (Read-only operation).

28 UC-27: Manage Users

28.1 Summary

This use case describes how an Admin manages all user accounts within the system. The Admin has the authority to view the user list, modify information, assign Roles (Student, Teacher, or Admin), and lock or delete accounts to maintain the security and integrity of the online examination system.

28.2 Flow of Events

28.2.1 Basic Flow

1. Admin accesses the "User Management" module from the Admin Dashboard.
2. The system queries and displays a list of all users, including: Username, Email, Full Name, Role, and Status (Active/Locked).
3. Admin performs one of the following administrative actions:

- Add New User: Admin enters identification details and assigns a specific role directly.
- Edit Information: Admin updates the full name or changes the Role (e.g., promoting a Student to a Teacher).
- Lock/Unlock Account: Admin toggles the account status. A locked account will be denied access during UC-02: Log In.
- Delete Account: Admin selects and removes obsolete or invalid accounts.

4. The system requests confirmation for critical actions (Delete, Lock).
5. The system updates the changes in the Users table in the database.
6. The system displays a notification: "User updated successfully".

28.2.2 Alternative Flows

- Search and Filter: Admin can search for users by Username/Email or filter by Role for efficient management.
- Self-Deletion Prevention: If the Admin attempts to delete their currently logged-in account, the system rejects the request with an error: "You cannot delete your own administrative account" (BR-02 of FR-10).
- Data Duplication: If adding a new user with an existing Email/Username, the system displays: "Email/Username already exists" (BR-01, BR-02 or FR-01).

28.3 Special Requirements

- Security: Only accounts with the Admin role are authorized to access and execute this use case (FR-11).
- Data Integrity: When deleting a user, the system must handle foreign key constraints (e.g., deleting related attempts or anonymizing data) to prevent report errors.
- Performance: API response time for retrieving or updating user data must be $\leq 500ms\ for\ \leq 20$ concurrent users (NFR-01).

28.4 Pre-conditions

- The user is successfully logged in as an Admin.
- The system is operational and connected to the database.

28.5 Post-conditions

- User account information is updated in the database.
- User access rights are modified or terminated immediately based on the Admin's actions.

28.6 Extension Points

- None.

29 UC-28: Manage Roles and Permissions

29.1 Summary

This use case describes how an Admin manages system Roles and their corresponding Permissions. The Admin can define what actions each role (e.g., Teacher) is authorized to perform (such as *Create Quiz* or *Delete Question*) to ensure system security and access control (RBAC).

29.2 Flow of Events

29.2.1 Basic Flow

1. Admin accesses the "Roles & Permissions" module from the Admin Dashboard.
2. The system displays a list of existing roles (Admin, Teacher, Student) along with their current assigned permissions.
3. Admin selects a role to edit or clicks "Add New Role".

4. The system displays a Permission Matrix, including functional modules such as: *User Management, Quiz Management, Result Access, and Question Bank*.

5. Admin performs the following:

   - Checks to grant a permission.

   - Unchecks to revoke a permission.

6. Admin clicks the "Save Permissions" button.

7. The system performs a logic check: Ensures the Admin role always retains full administrative privileges to prevent a system "lock-out" scenario.

8. The system updates the Roles and Permissions tables (via the Role_Permissions junction table) in the database.

9. The system displays a notification: "Permissions updated successfully".

29.2.2 Alternative Flows

- Create New Role: Admin enters a new role name (e.g., "Assistant") and begins assigning permissions from scratch.

- Delete Role: Admin can delete custom roles, excluding system default roles. If a role is currently assigned to users, the system displays: "Cannot delete role while it is assigned to users".

- Access Denied: If a user attempts to access a feature for which their role's permission was recently revoked, the system blocks the request and displays: "403 Forbidden - You don't have permission to access this resource".

29.3 Special Requirements

Immediate Effect: Upon saving changes, user permissions must update immediately (e.g., by refreshing the session or clearing authorization cache).

Data Integrity: The system must maintain at least one Admin account with full permissions at all times.

Performance:

- API response time for retrieving or updating permissions must be $\leq 500ms\ for\ \leq 20$ concurrent users.
- General system operations must ensure a smooth experience during the demo.

29.4 Pre-conditions

- The user is logged in with the Admin role.
- The functional permission categories are already defined in the database.

29.5 Post-conditions

- The permission mapping in the database is updated.
- User actions throughout the system are strictly controlled by the newly established permission set.

29.6 Extension Points

- None

30 UC-29: Purge Data

30.1 Summary

This use case describes how an Admin performs deep cleaning of redundant data, temporary records, or obsolete information (such as activity logs, old quiz attempts, and soft-deleted records) from the system. The goal is to reclaim storage space and optimize database query performance.

30.2 Flow of Events

30.2.1 Basic Flow

1. Admin accesses the "System Maintenance" module from the Admin Dashboard.
2. The system displays clean-up categories with estimated storage sizes:

- System Logs: Old activity and error logs.
- Old Attempts: Quiz attempts from previous academic years.
- Soft-deleted Records: Records marked as deleted but still physically present in the DB.

3. Admin selects the categories and sets filters (e.g., "Data prior to 01/01/2025").
4. Admin clicks "Analyze" to calculate the exact number of records to be permanently removed.
5. The system displays a final confirmation summary.
6. Admin confirms the Purge (requires re-authentication with Admin password for security).
7. The system executes a Hard Delete command on the corresponding tables in the database.
8. The system displays a notification: "Data purged successfully. [X] records removed."

30.2.2 Alternative Flows

- Operation Cancelled: If the Admin selects "Cancel" at any step, the system aborts the process without deleting data.
- Locked Data: If specific records are currently in use (e.g., a student is currently taking a quiz), the system skips those records and provides a post-process report.
- Batch Processing (NFR-01): If the volume of data is too large, the system processes the deletion in batches to prevent database timeouts or interface freezing.

30.3 Special Requirements

- Irreversibility: Data removed via Purge is permanently deleted. The system must strongly advise the Admin to perform a Backup before execution.

- Data Safety: Only records without active Foreign Key constraints or those meeting specific business logic can be purged to prevent database corruption.
- Performance (NFR-01):
- Initiation and status API responses must be $\leq 500ms$.
- The actual heavy deletion process should ideally be scheduled during low-traffic hours.

30.4 Pre-conditions

- The user is logged in with the Admin role.
- A recent database backup has been successfully completed.

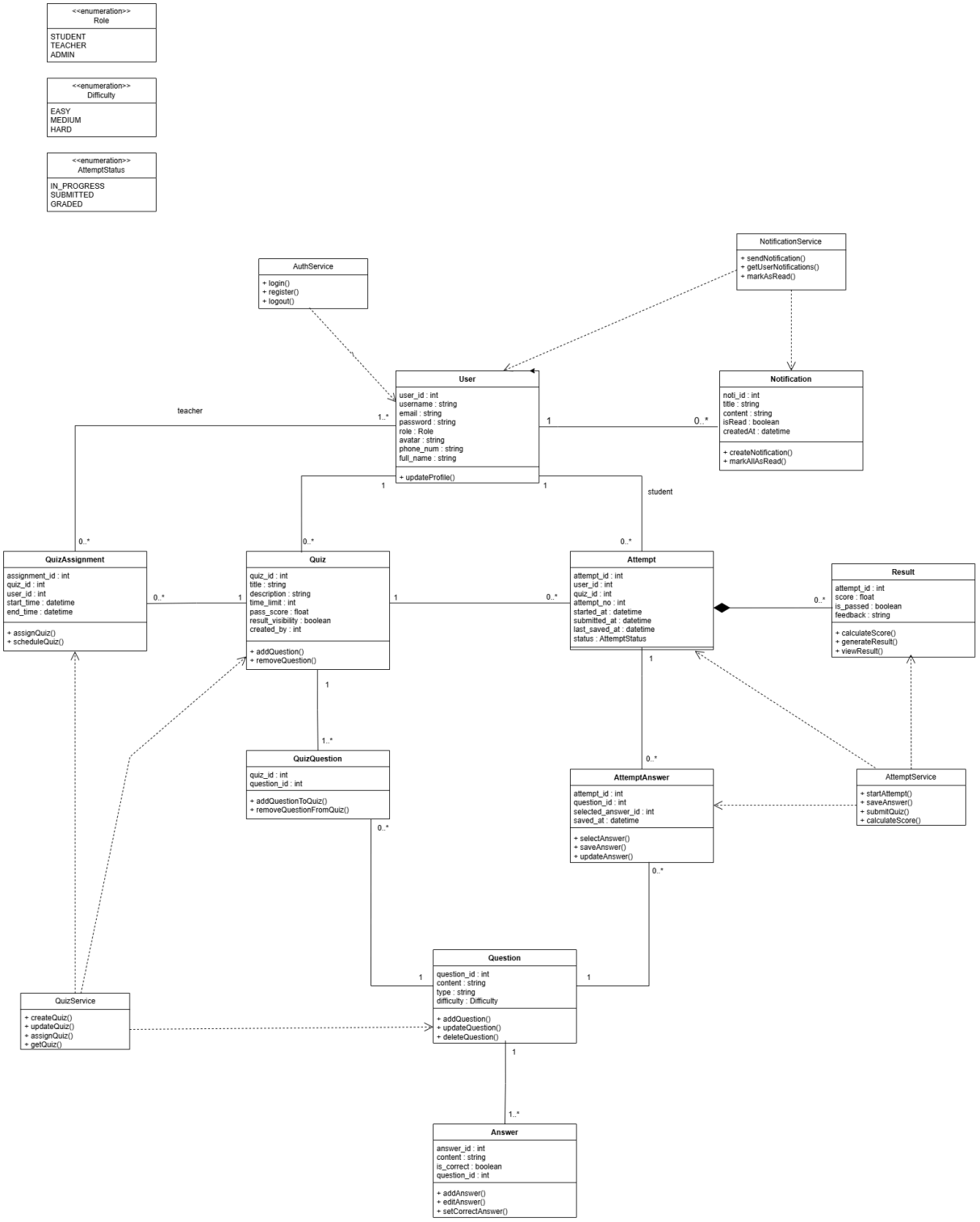
30.5 Post-conditions

- Database storage usage is significantly reduced.
- Query performance on primary tables (Attempts, Questions) is improved due to reduced index sizes.

30.6 Extension Points

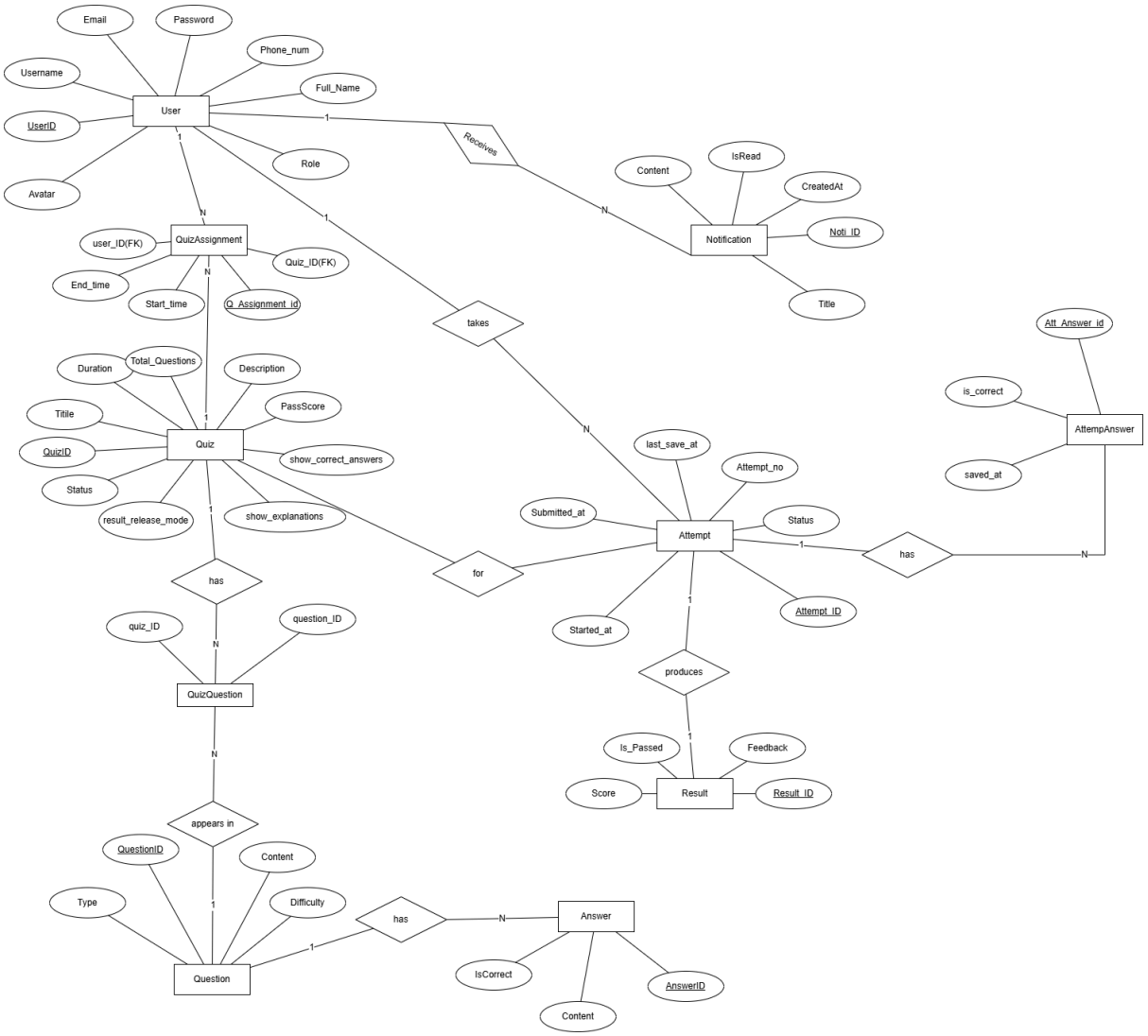
- Scheduled Purge: Allows the Admin to set a recurring maintenance schedule (e.g., every 6 months) without manual intervention.

V. CLASS DIAGRAM

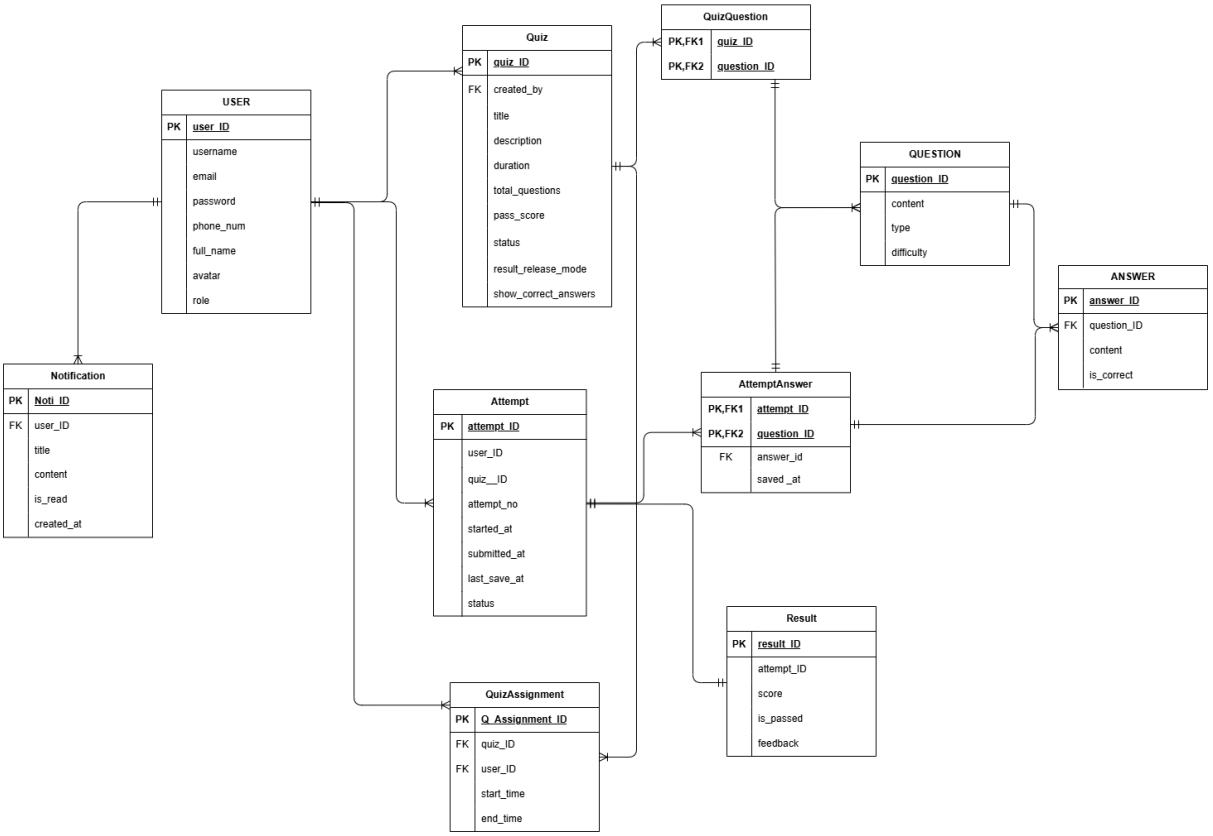


VI. Data Model:

1. concept levels

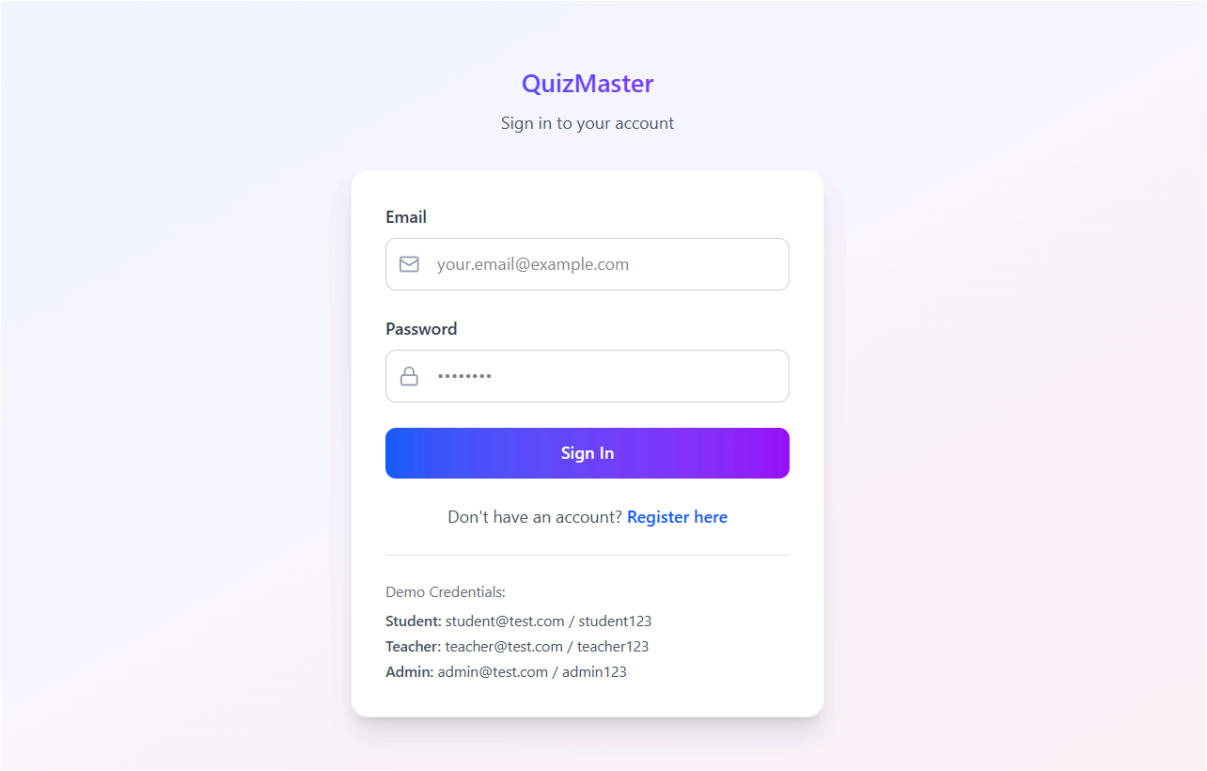


2. logical level

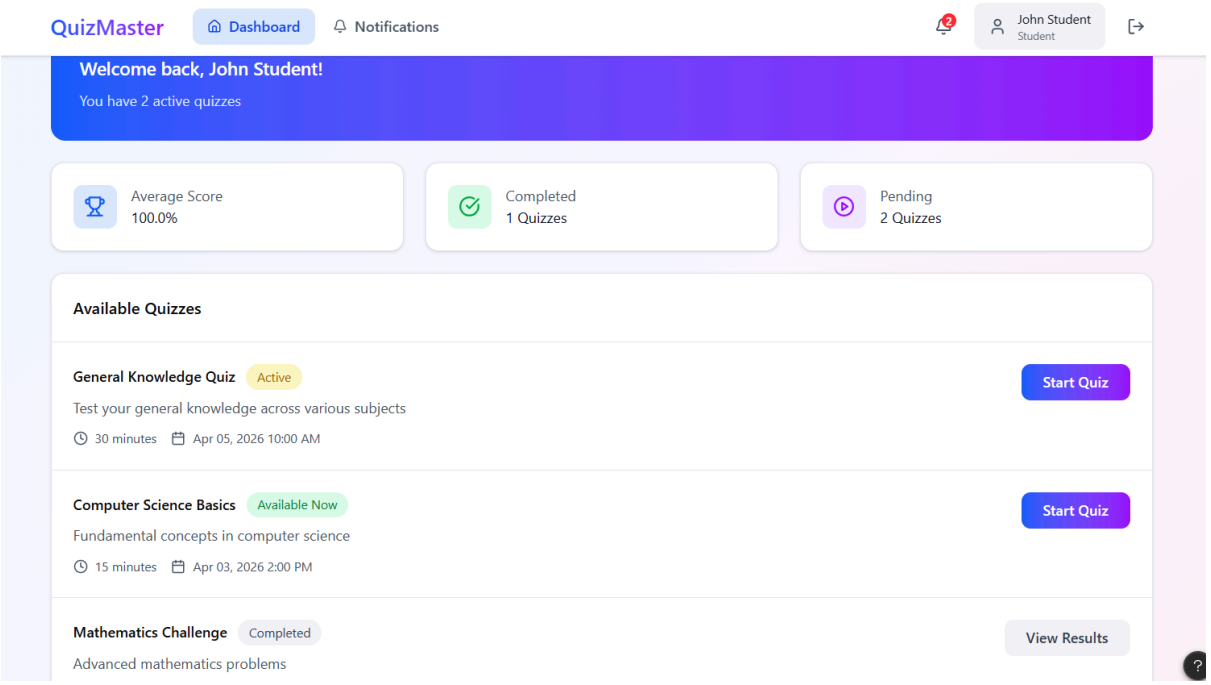


VII. Interface Design Description:

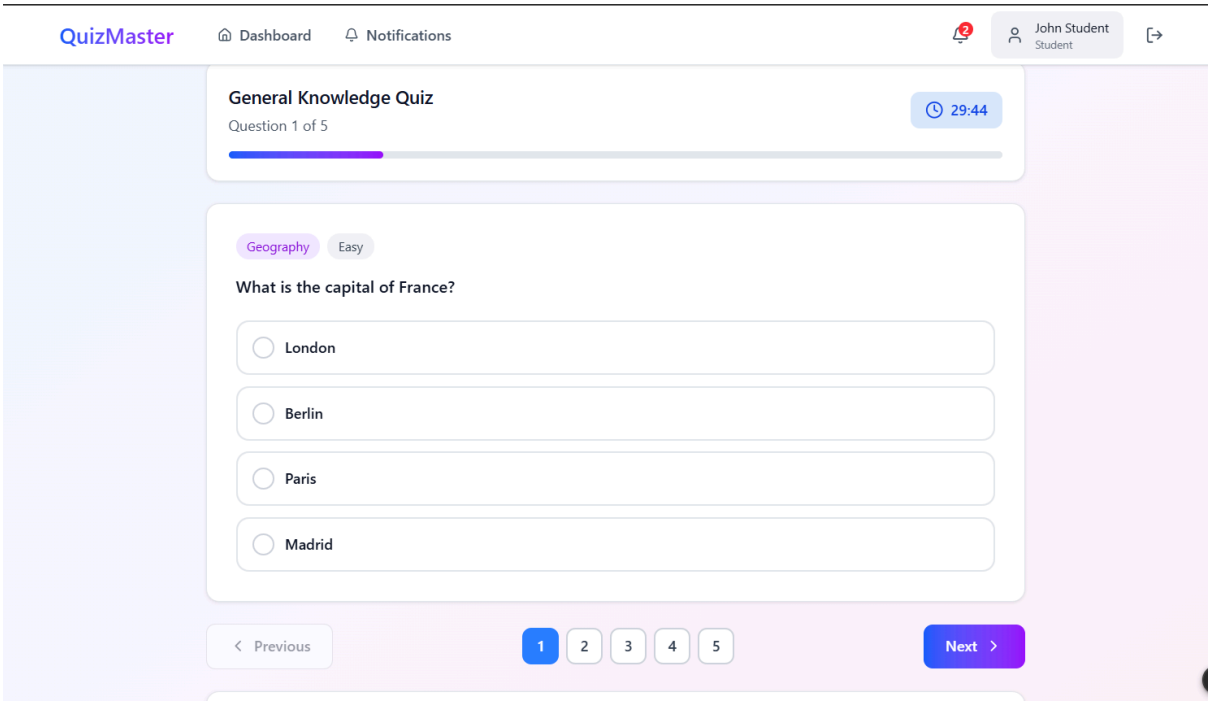
1. register screen

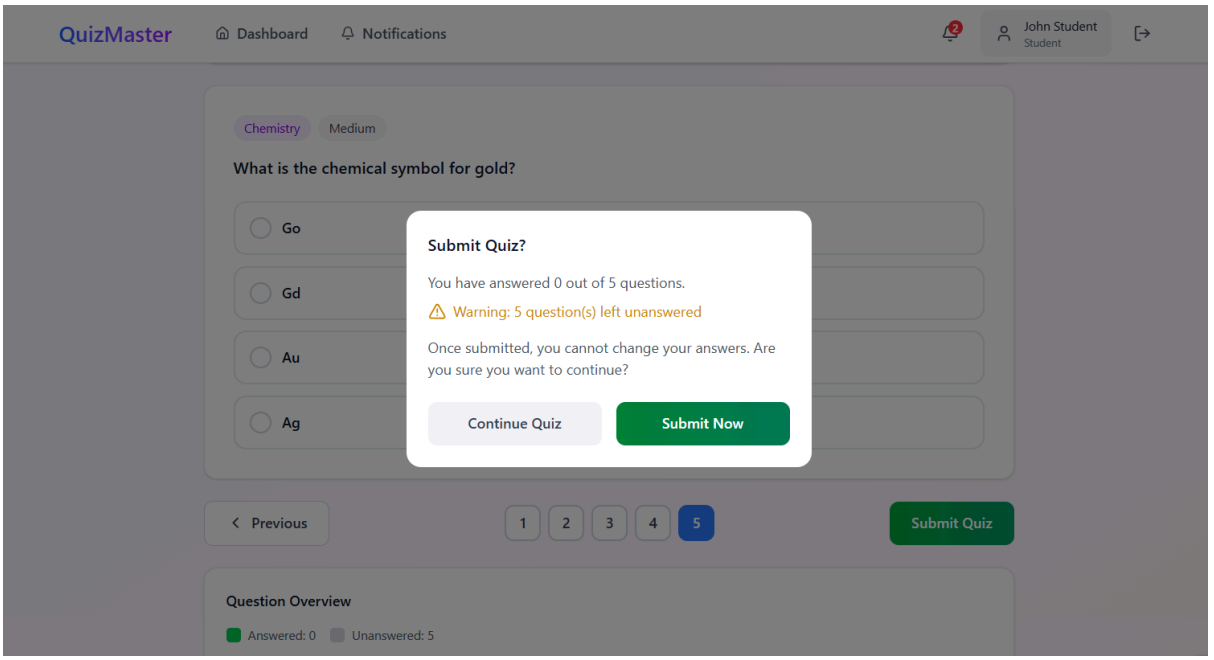


2. Main Screen – Student Role (After Login)

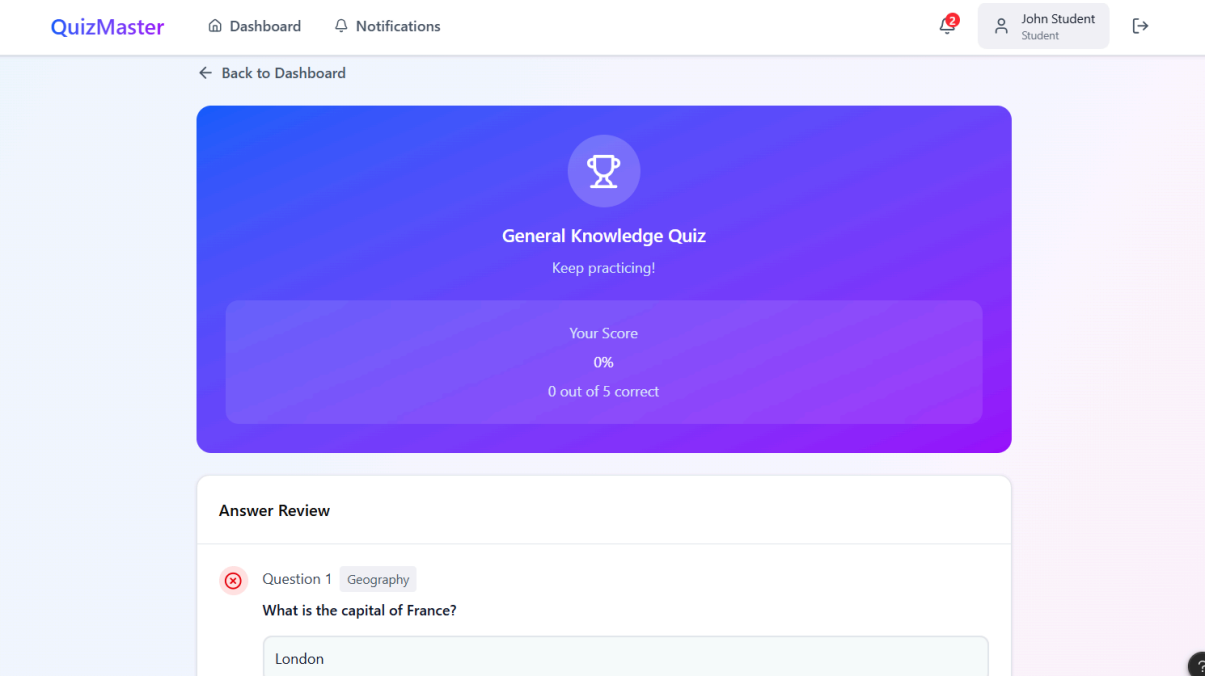


3. Quiz Taking Screen – Student Role

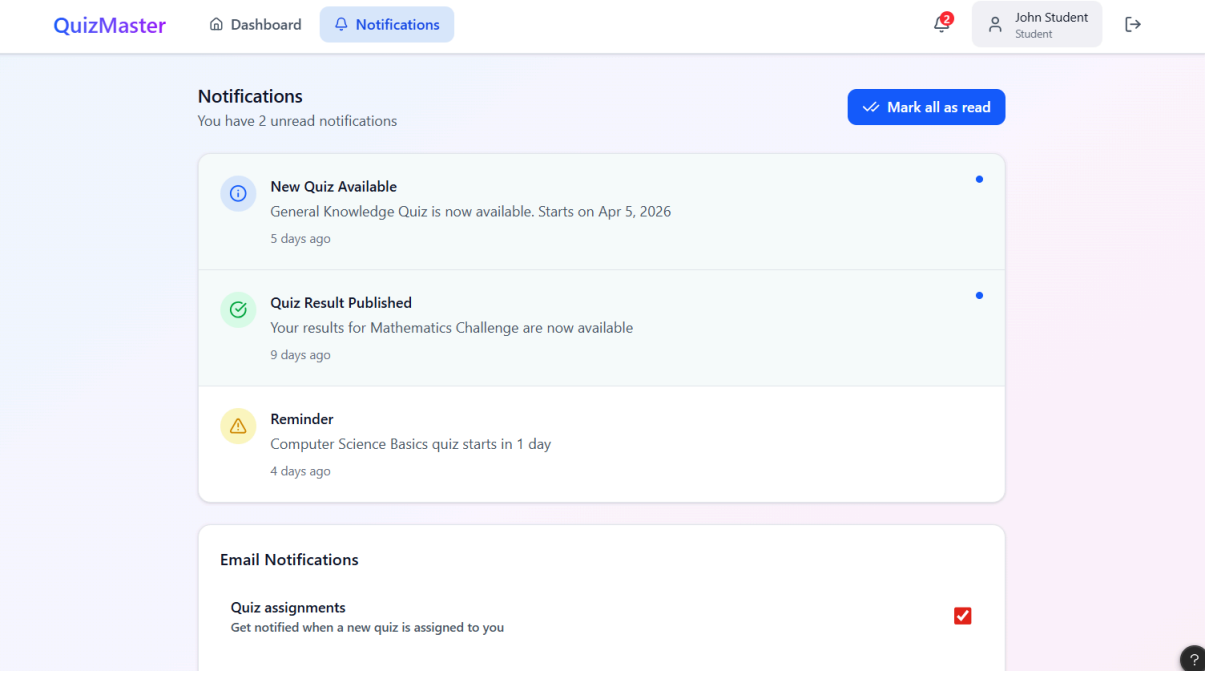




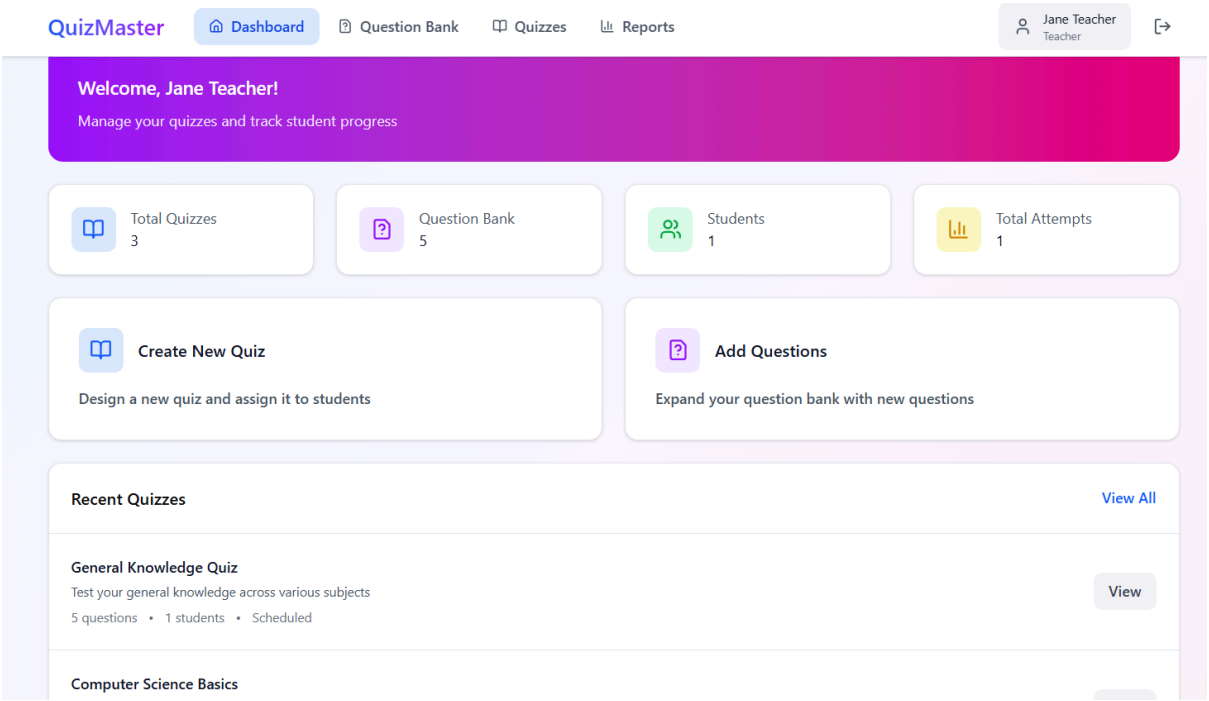
4. Quiz Result Screen – Student Role



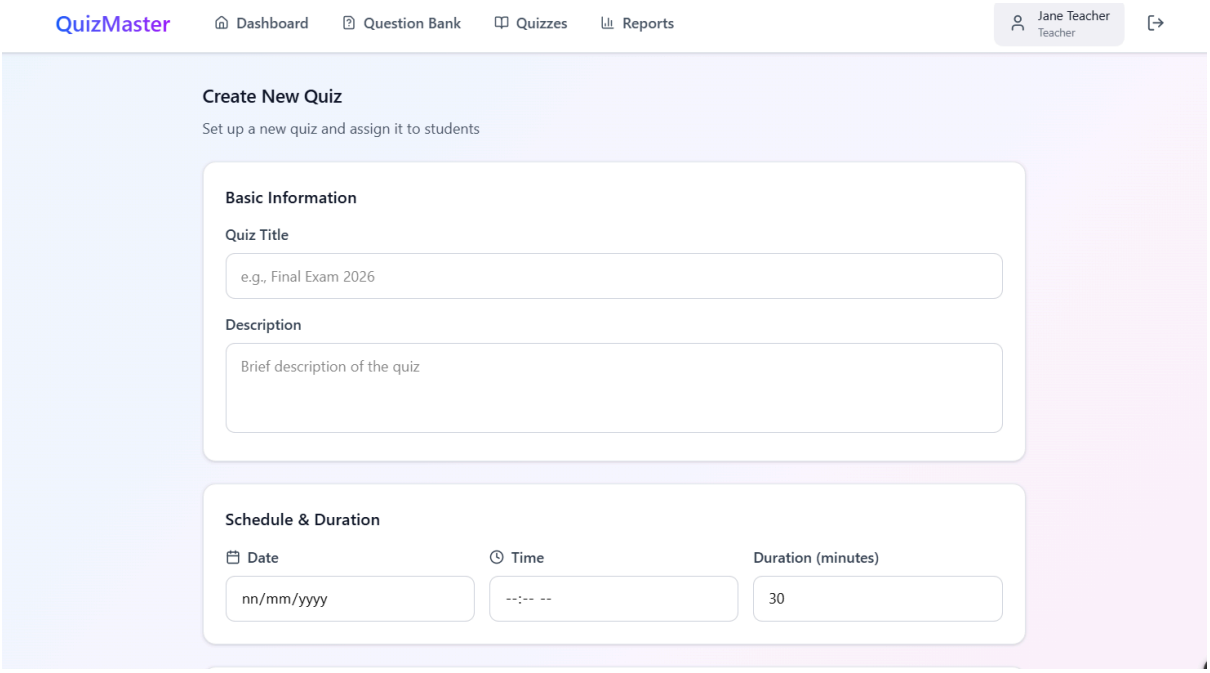
5. Notification Screen



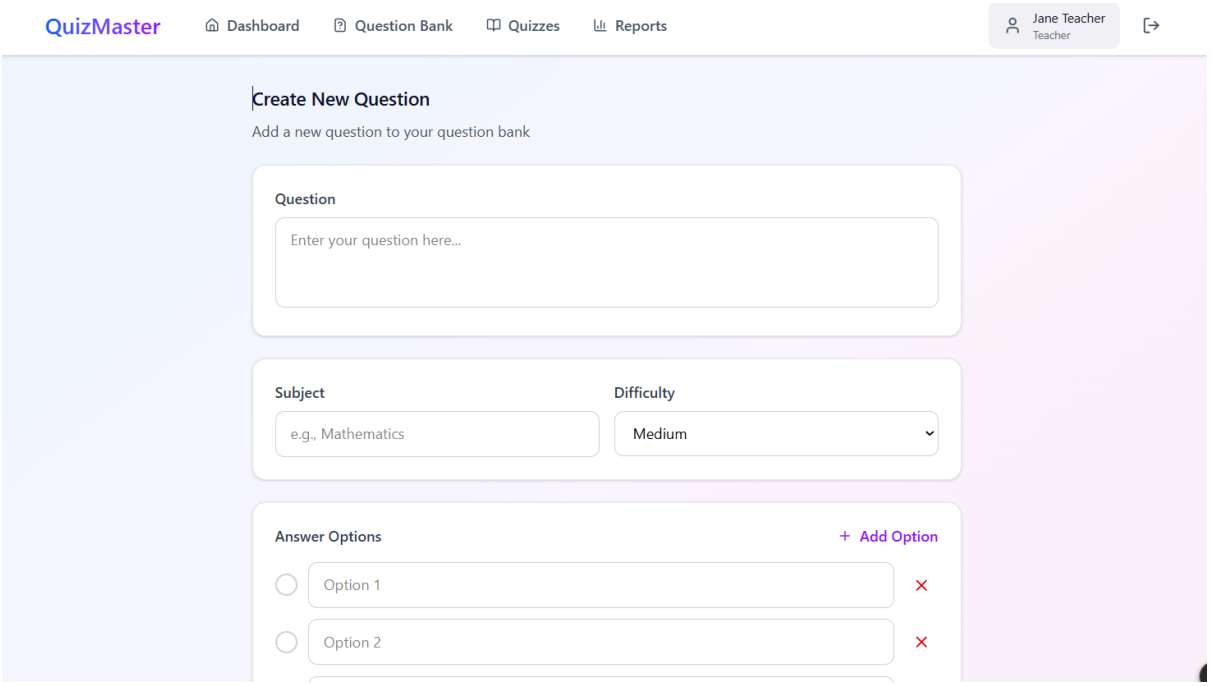
6. Main Screen – Teacher Role (After Login)



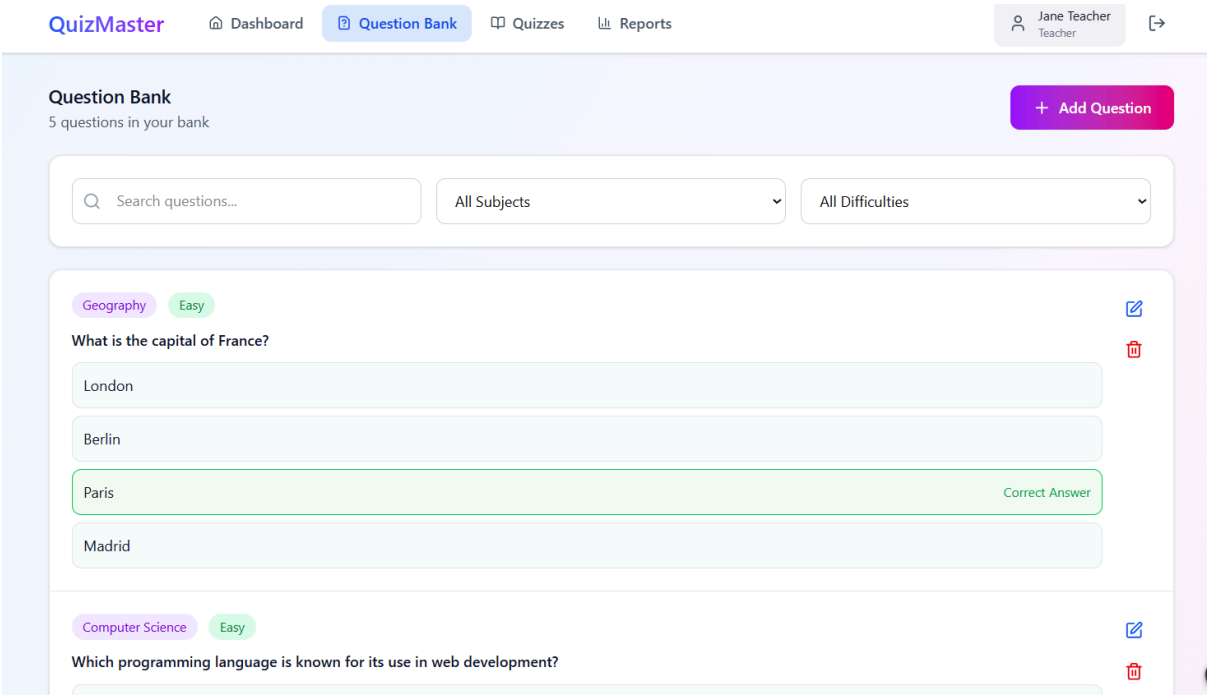
7. Create Quiz Screen – Teacher Role



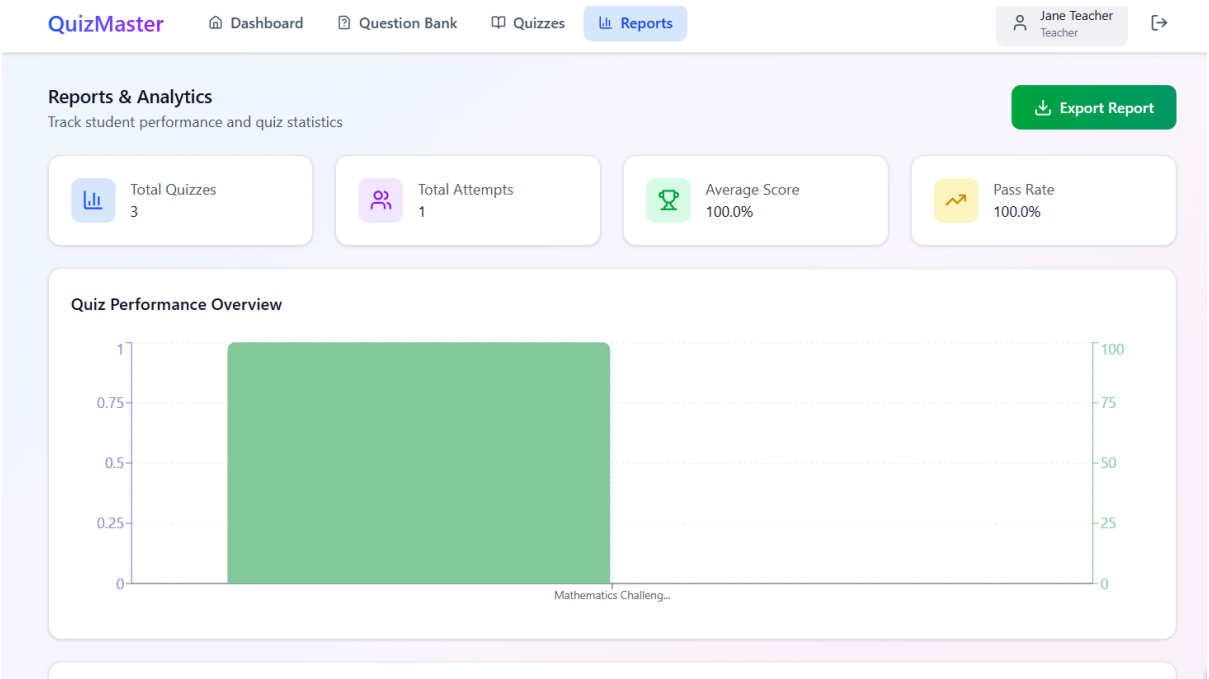
8. Add Question to Question Bank – Teacher Role



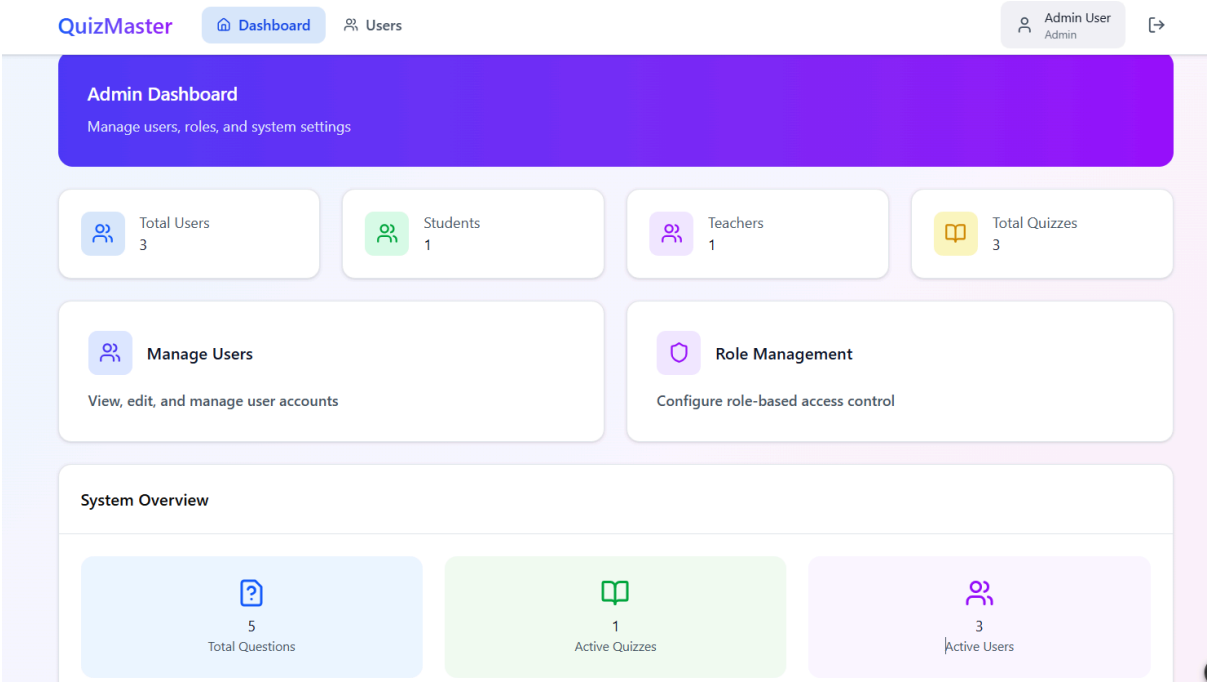
9. Question Bank

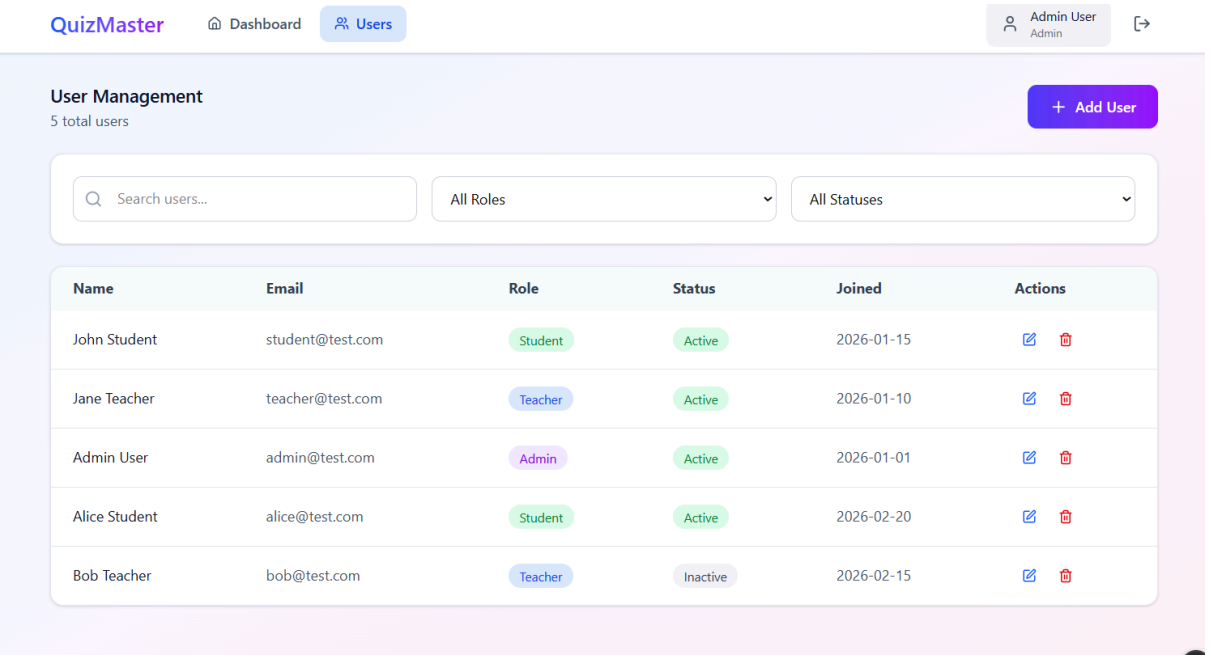


10. View result-Role teacher



11. Main Screen – Admin Role (After Login)



12. User Management Screen – Admin Role

Quiz examinations System	version: 1.1
System specification	Date: 6/4/2026